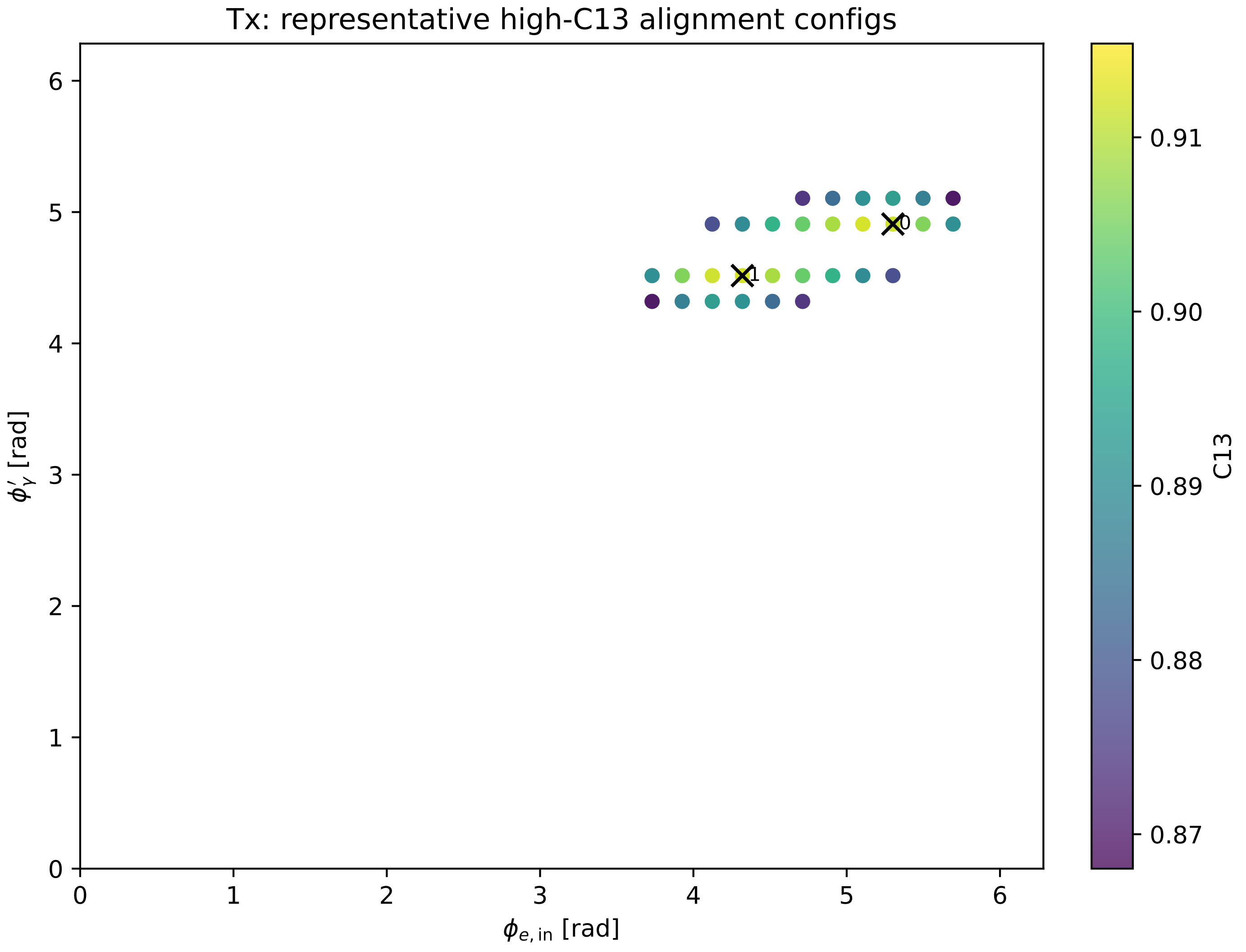
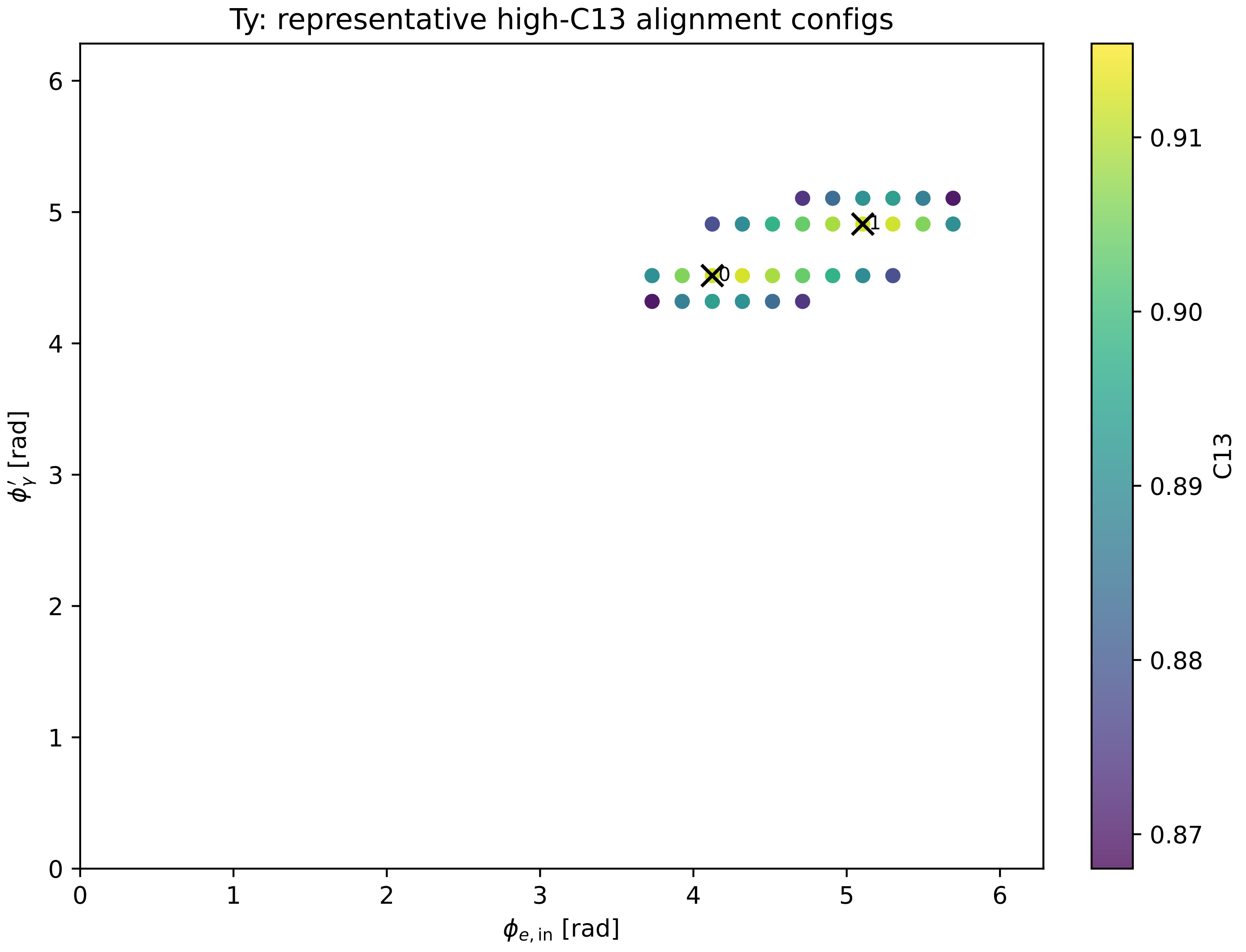
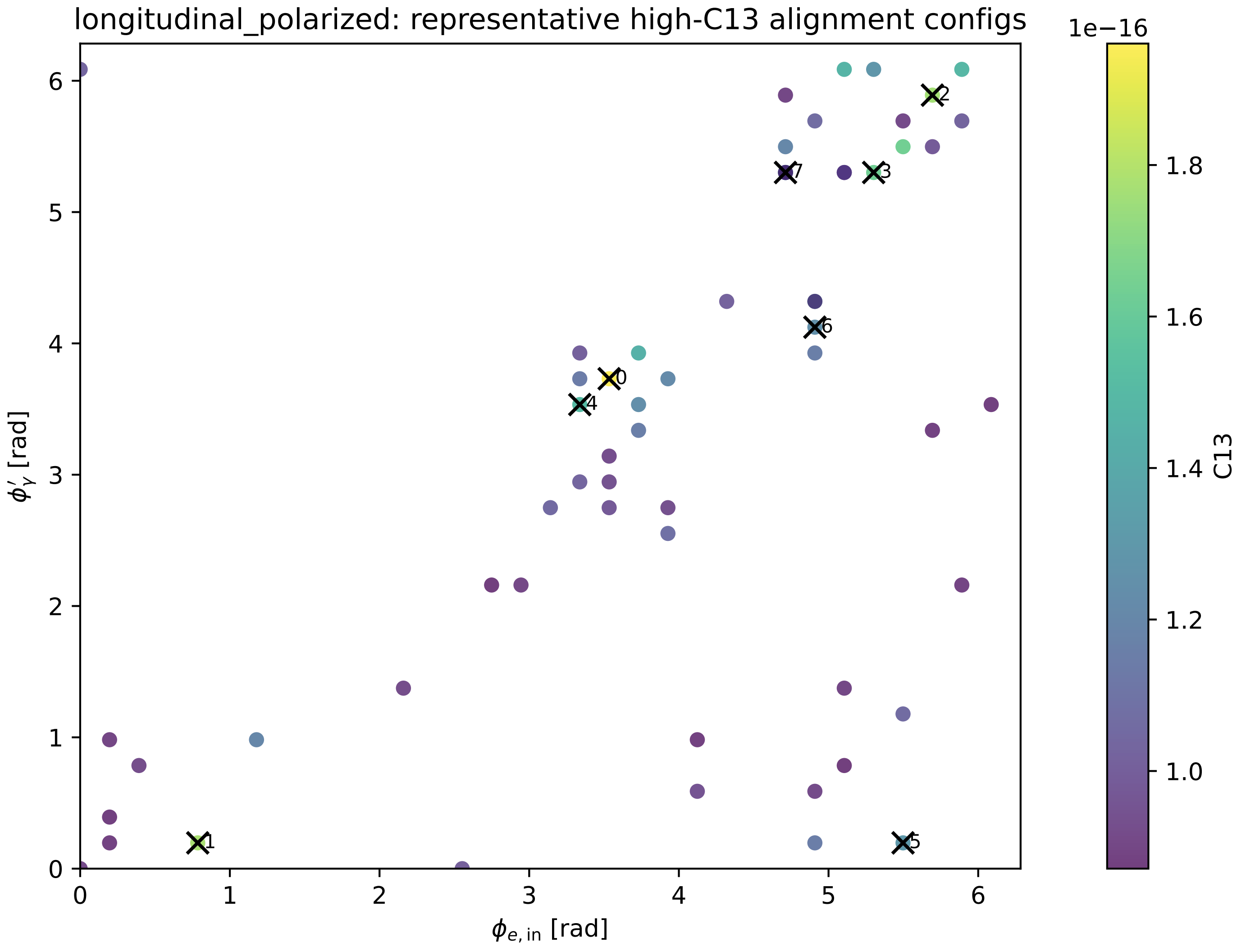
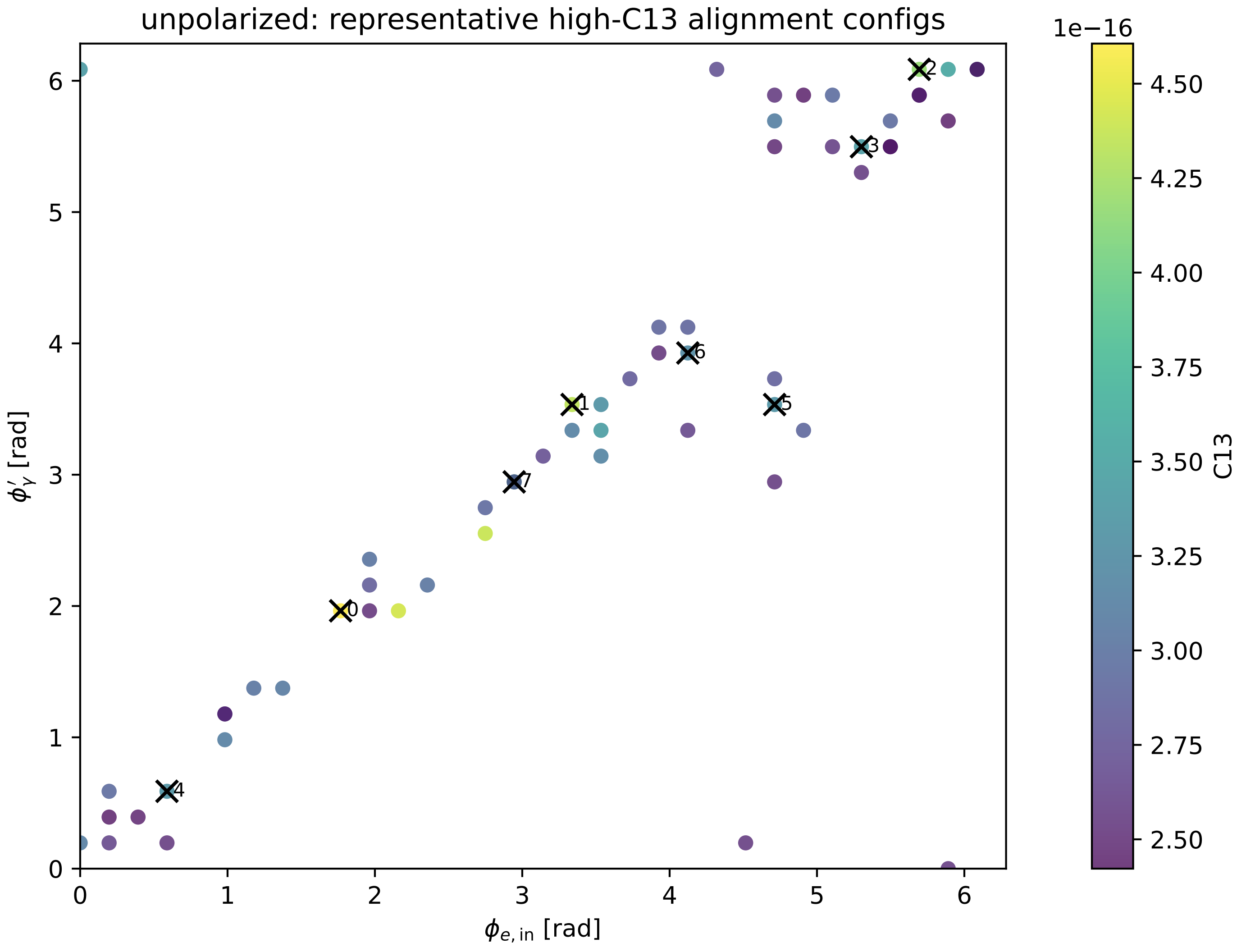

Tx: representative high-C13 alignment configs
ϕ'_γ [rad]
$\phi_{e,\mathrm{in}}$ [rad]
0
1
C13
0.87
0.88
0.89
0.90
0.91



Ty: representative high-C13 alignment configs
ϕ'_{γ} [rad]
$\phi_{e,\mathrm{in}}$ [rad]
0
1
0
1
2
3
4
5
6
C13
0.87
0.88
0.89
0.90
0.91

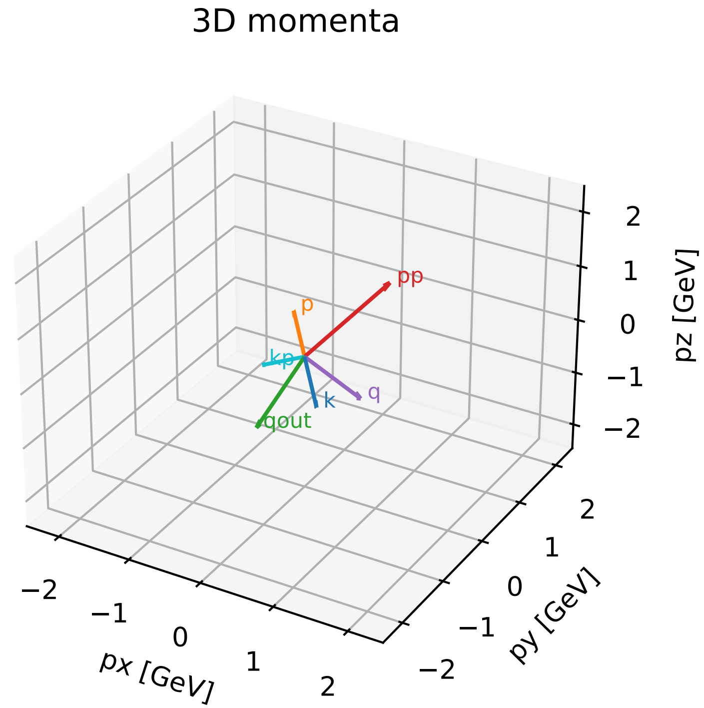


longitudinal_polarized: representative high-C13 alignment configs
ϕ'_γ [rad]
$\phi_{e,\mathrm{in}}$ [rad]
C13
1e−16
0
1
2
3
4
5
6
1.0
1.2
1.4
1.6
1.8

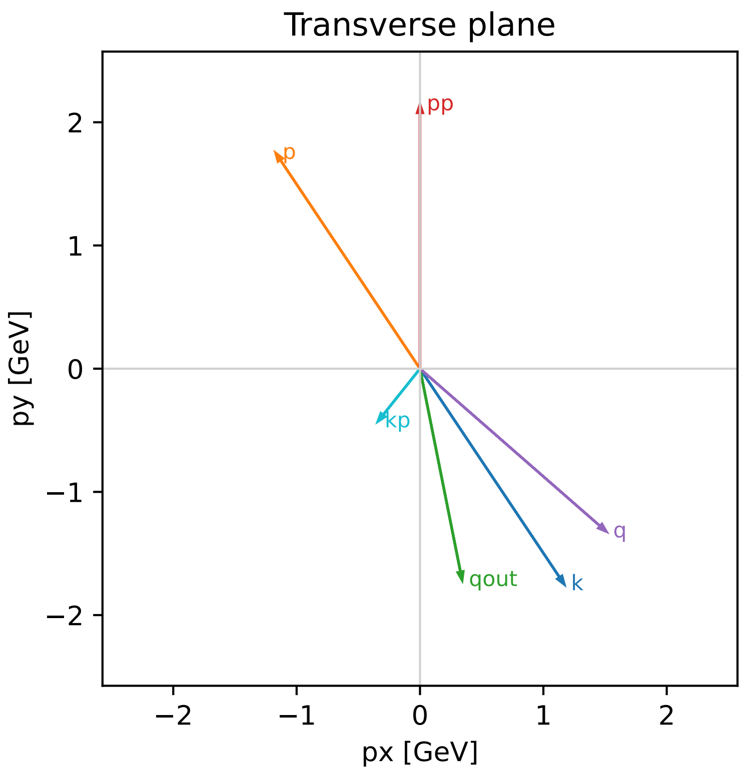


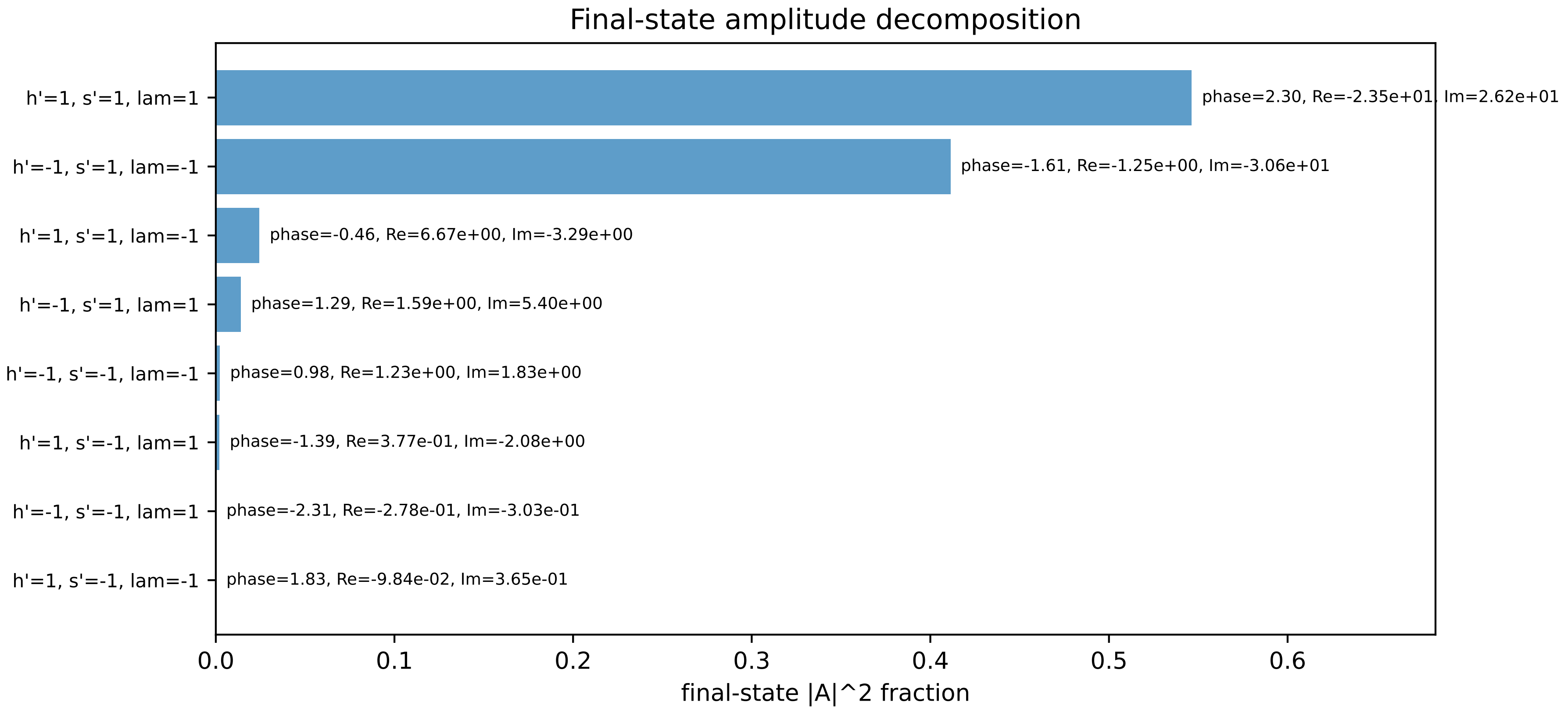
unpolarized: representative high-C13 alignment configs
ϕ'_{γ} [rad]
$\phi_{e,\mathrm{in}}$ [rad]
C13
1e−16
4.50
4.25
4.00
3.75
3.50
3.25
3.00
2.75
2.50
0
1
2
3
4
5
6
7

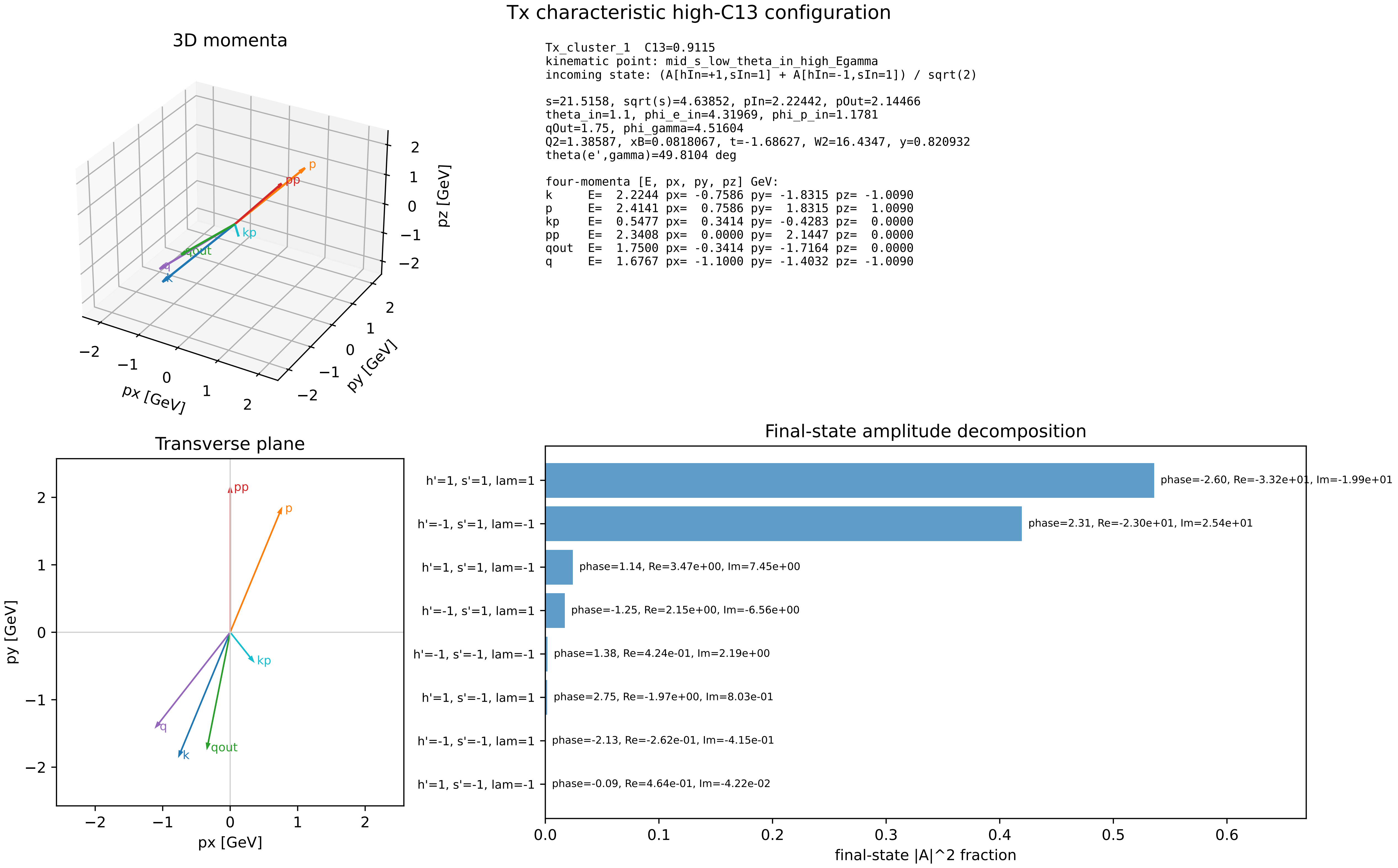
Tx characteristic high-C13 configuration

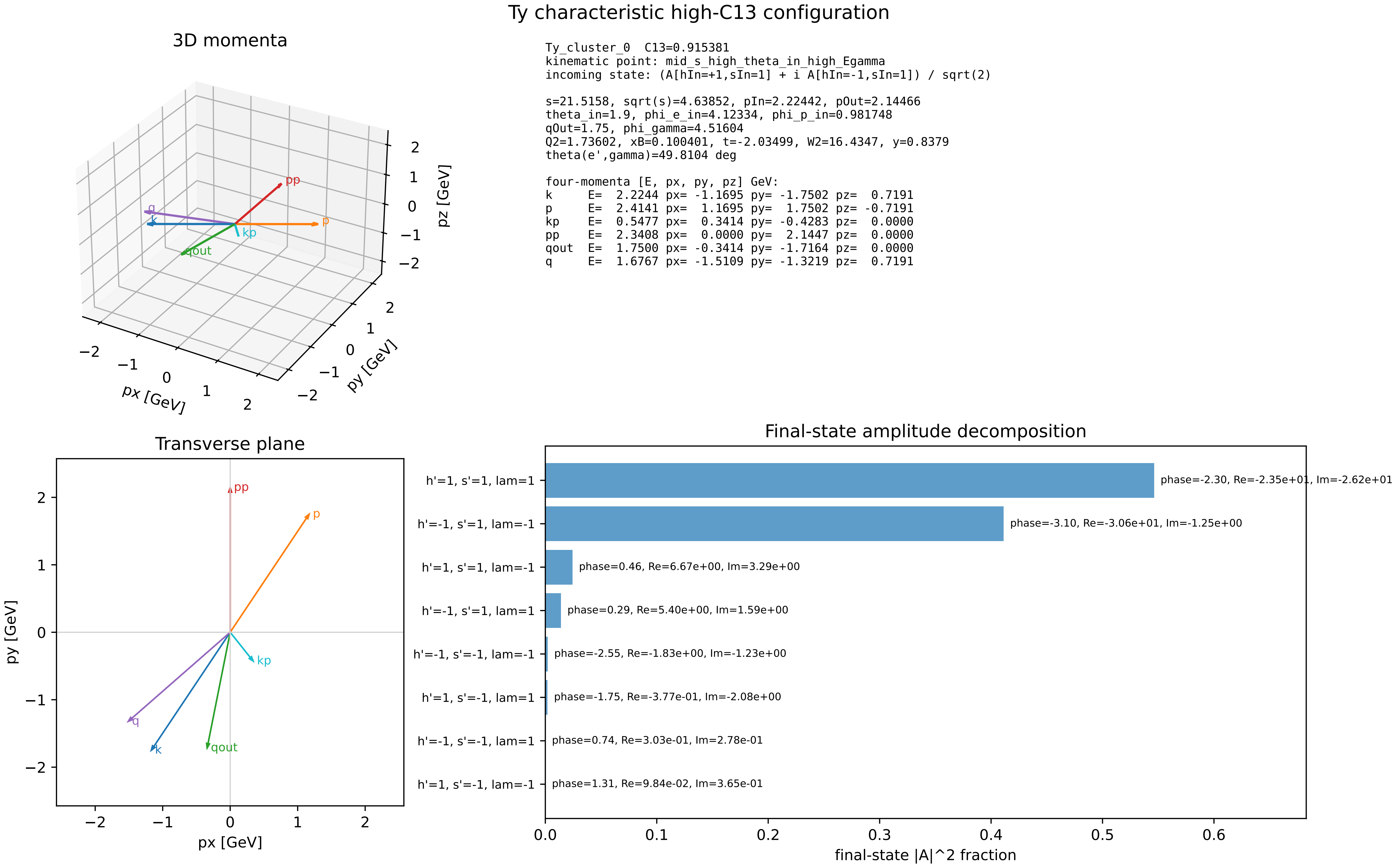


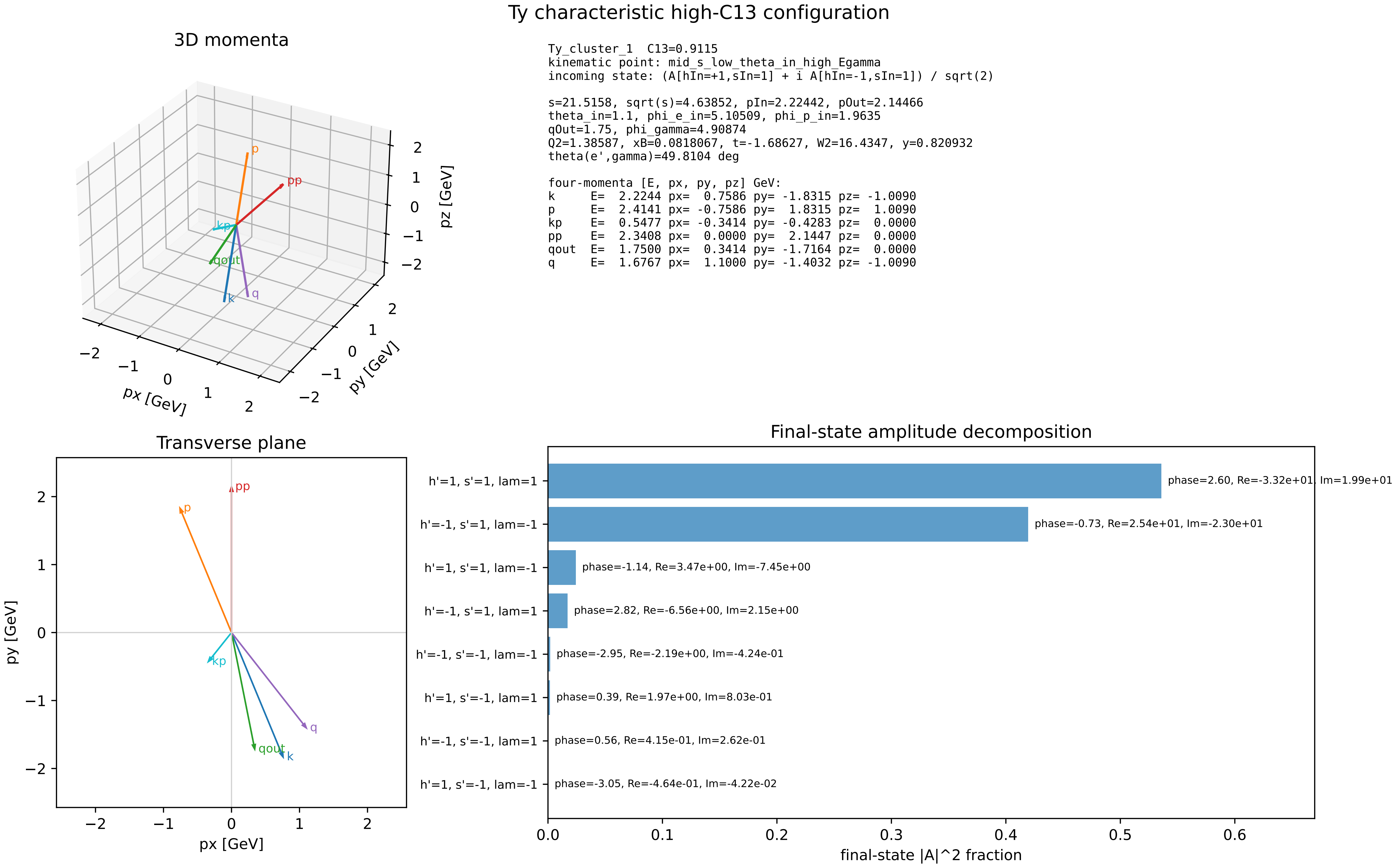

```
Tx_cluster_0  C13=0.915381
kinematic point: mid_s_high_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] + A[hIn=-1,sIn=1]) / sqrt(2)

s=21.5158, sqrt(s)=4.63852, pIn=2.22442, pOut=2.14466
theta_in=1.9, phi_e_in=5.30144, phi_p_in=2.15984
qOut=1.75, phi_gamma=4.90874
Q2=1.73602, xB=0.100401, t=-2.03499, W2=16.4347, y=0.8379
theta(e',gamma)=49.8104 deg

four-momenta [E, px, py, pz] GeV:
k     E=  2.2244 px=  1.1695 py= -1.7502 pz=  0.7191
p     E=  2.4141 px= -1.1695 py=  1.7502 pz= -0.7191
kp    E=  0.5477 px= -0.3414 py= -0.4283 pz=  0.0000
pp    E=  2.3408 px=  0.0000 py=  2.1447 pz=  0.0000
qout  E=  1.7500 px=  0.3414 py= -1.7164 pz=  0.0000
q     E=  1.6767 px=  1.5109 py= -1.3219 pz=  0.7191
```



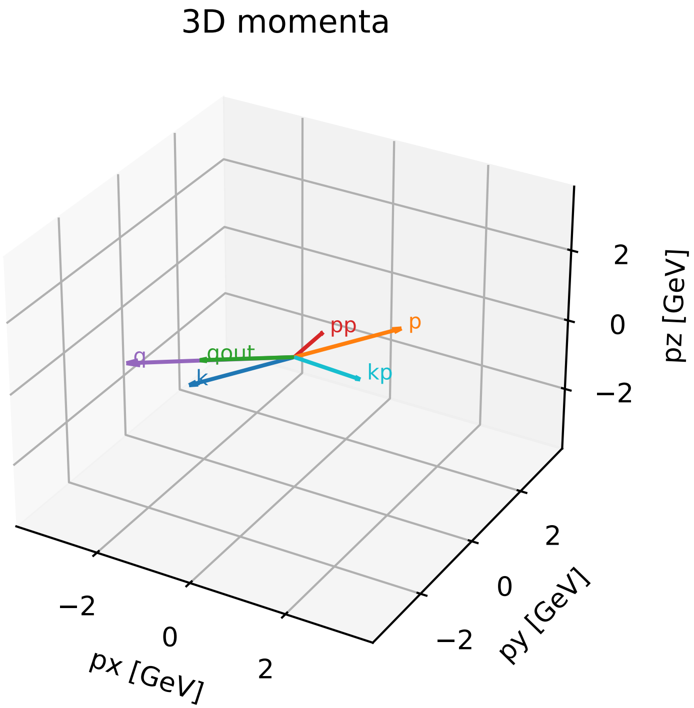



Tx characteristic high-C13 configuration
3D momenta
px [GeV]
py [GeV]
pz [GeV]
Tx_cluster_1 C13=0.9115
kinematic point: mid_s_low_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] + A[hIn=-1,sIn=1]) / sqrt(2)
s=21.5158, sqrt(s)=4.63852, pIn=2.22442, pOut=2.14466
theta_in=1.1, phi_e_in=4.31969, phi_p_in=1.1781
qOut=1.75, phi_gamma=4.51604
Q2=1.38587, xB=0.0818067, t=-1.68627, W2=16.4347, y=0.820932
theta(e',gamma)=49.8104 deg
four-momenta [E, px, py, pz] GeV:
k   E= 2.2244 px= -0.7586 py= -1.8315 pz= -1.0090
p   E= 2.4141 px= 0.7586 py= 1.8315 pz= 1.0090
kp   E= 0.5477 px= 0.3414 py= -0.4283 pz= 0.0000
pp   E= 2.3408 px= 0.0000 py= 2.1447 pz= 0.0000
qout E= 1.7500 px= -0.3414 py= -1.7164 pz= 0.0000
q   E= 1.6767 px= -1.1000 py= -1.4032 pz= -1.0090
Transverse plane
px [GeV]
py [GeV]
Final-state amplitude decomposition
h'=1, s'=1, lam=1
phase=-2.60, Re=-3.32e+01, Im=-1.99e+01
h'=-1, s'=1, lam=-1
phase=2.31, Re=-2.30e+01, Im=2.54e+01
h'=1, s'=1, lam=-1
phase=1.14, Re=3.47e+00, Im=7.45e+00
h'=-1, s'=1, lam=1
phase=-1.25, Re=2.15e+00, Im=-6.56e+00
h'=-1, s'=-1, lam=-1
phase=1.38, Re=4.24e-01, Im=2.19e+00
h'=1, s'=-1, lam=1
phase=2.75, Re=-1.97e+00, Im=8.03e-01
h'=-1, s'=-1, lam=1
phase=-2.13, Re=-2.62e-01, Im=-4.15e-01
h'=1, s'=-1, lam=-1
phase=-0.09, Re=4.64e-01, Im=-4.22e-02
final-state |A|^2 fraction

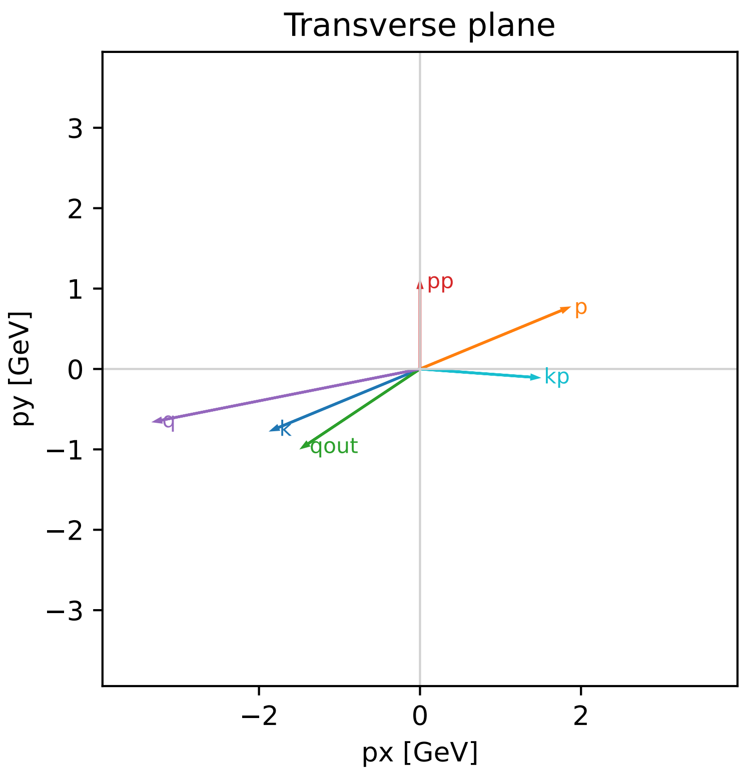

Ty characteristic high-C13 configuration
3D momenta
pp
p
q
k
kp
qout
px [GeV]
py [GeV]
pz [GeV]
Ty_cluster_0 C13=0.915381
kinematic point: mid_s_high_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] + i A[hIn=-1,sIn=1]) / sqrt(2)
s=21.5158, sqrt(s)=4.63852, pIn=2.22442, pOut=2.14466
theta_in=1.9, phi_e_in=4.12334, phi_p_in=0.981748
qOut=1.75, phi_gamma=4.51604
Q2=1.73602, xB=0.100401, t=-2.03499, W2=16.4347, y=0.8379
theta(e',gamma)=49.8104 deg
four-momenta [E, px, py, pz] GeV:
k   E= 2.2244 px= -1.1695 py= -1.7502 pz= 0.7191
p   E= 2.4141 px= 1.1695 py= 1.7502 pz= -0.7191
kp   E= 0.5477 px= 0.3414 py= -0.4283 pz= 0.0000
pp   E= 2.3408 px= 0.0000 py= 2.1447 pz= 0.0000
qout E= 1.7500 px= -0.3414 py= -1.7164 pz= 0.0000
q   E= 1.6767 px= -1.5109 py= -1.3219 pz= 0.7191
Transverse plane
px [GeV]
py [GeV]
Final-state amplitude decomposition
h'=1, s'=1, lam=1
phase=-2.30, Re=-2.35e+01, Im=-2.62e+01
h'=-1, s'=1, lam=-1
phase=-3.10, Re=-3.06e+01, Im=-1.25e+00
h'=1, s'=1, lam=-1
phase=0.46, Re=6.67e+00, Im=3.29e+00
h'=-1, s'=1, lam=1
phase=0.29, Re=5.40e+00, Im=1.59e+00
h'=-1, s'=-1, lam=-1
phase=-2.55, Re=-1.83e+00, Im=-1.23e+00
h'=1, s'=-1, lam=1
phase=-1.75, Re=-3.77e-01, Im=-2.08e+00
h'=-1, s'=-1, lam=1
phase=0.74, Re=3.03e-01, Im=2.78e-01
h'=1, s'=-1, lam=-1
phase=1.31, Re=9.84e-02, Im=3.65e-01
final-state |A|^2 fraction

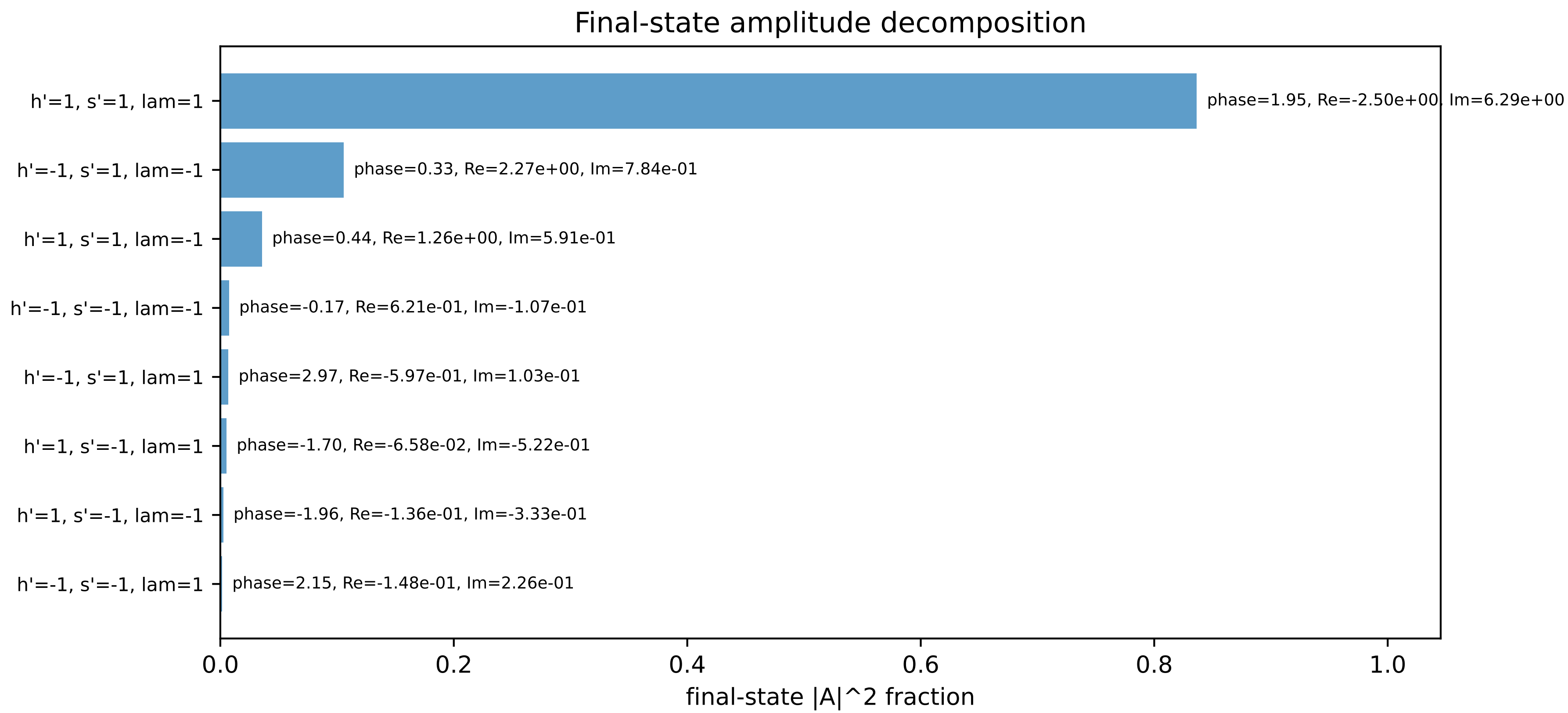

Ty characteristic high-C13 configuration
3D momenta
px [GeV]
py [GeV]
pz [GeV]
p
pp
kp
qout
k
q
Ty_cluster_1 C13=0.9115
kinematic point: mid_s_low_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] + i A[hIn=-1,sIn=1]) / sqrt(2)
s=21.5158, sqrt(s)=4.63852, pIn=2.22442, pOut=2.14466
theta_in=1.1, phi_e_in=5.10509, phi_p_in=1.9635
qOut=1.75, phi_gamma=4.90874
Q2=1.38587, xB=0.0818067, t=-1.68627, W2=16.4347, y=0.820932
theta(e',gamma)=49.8104 deg
four-momenta [E, px, py, pz] GeV:
k   E= 2.2244 px= 0.7586 py= -1.8315 pz= -1.0090
p   E= 2.4141 px= -0.7586 py= 1.8315 pz= 1.0090
kp   E= 0.5477 px= -0.3414 py= -0.4283 pz= 0.0000
pp   E= 2.3408 px= 0.0000 py= 2.1447 pz= 0.0000
qout E= 1.7500 px= 0.3414 py= -1.7164 pz= 0.0000
q   E= 1.6767 px= 1.1000 py= -1.4032 pz= -1.0090
Transverse plane
px [GeV]
py [GeV]
Final-state amplitude decomposition
h'=1, s'=1, lam=1
phase=2.60, Re=-3.32e+01, Im=1.99e+01
h'=-1, s'=1, lam=-1
phase=-0.73, Re=2.54e+01, Im=-2.30e+01
h'=1, s'=1, lam=-1
phase=-1.14, Re=3.47e+00, Im=-7.45e+00
h'=-1, s'=1, lam=1
phase=2.82, Re=-6.56e+00, Im=2.15e+00
h'=-1, s'=-1, lam=-1
phase=-2.95, Re=-2.19e+00, Im=-4.24e-01
h'=1, s'=-1, lam=1
phase=0.39, Re=1.97e+00, Im=8.03e-01
h'=-1, s'=-1, lam=1
phase=0.56, Re=4.15e-01, Im=2.62e-01
h'=1, s'=-1, lam=-1
phase=-3.05, Re=-4.64e-01, Im=-4.22e-02
final-state |A|^2 fraction

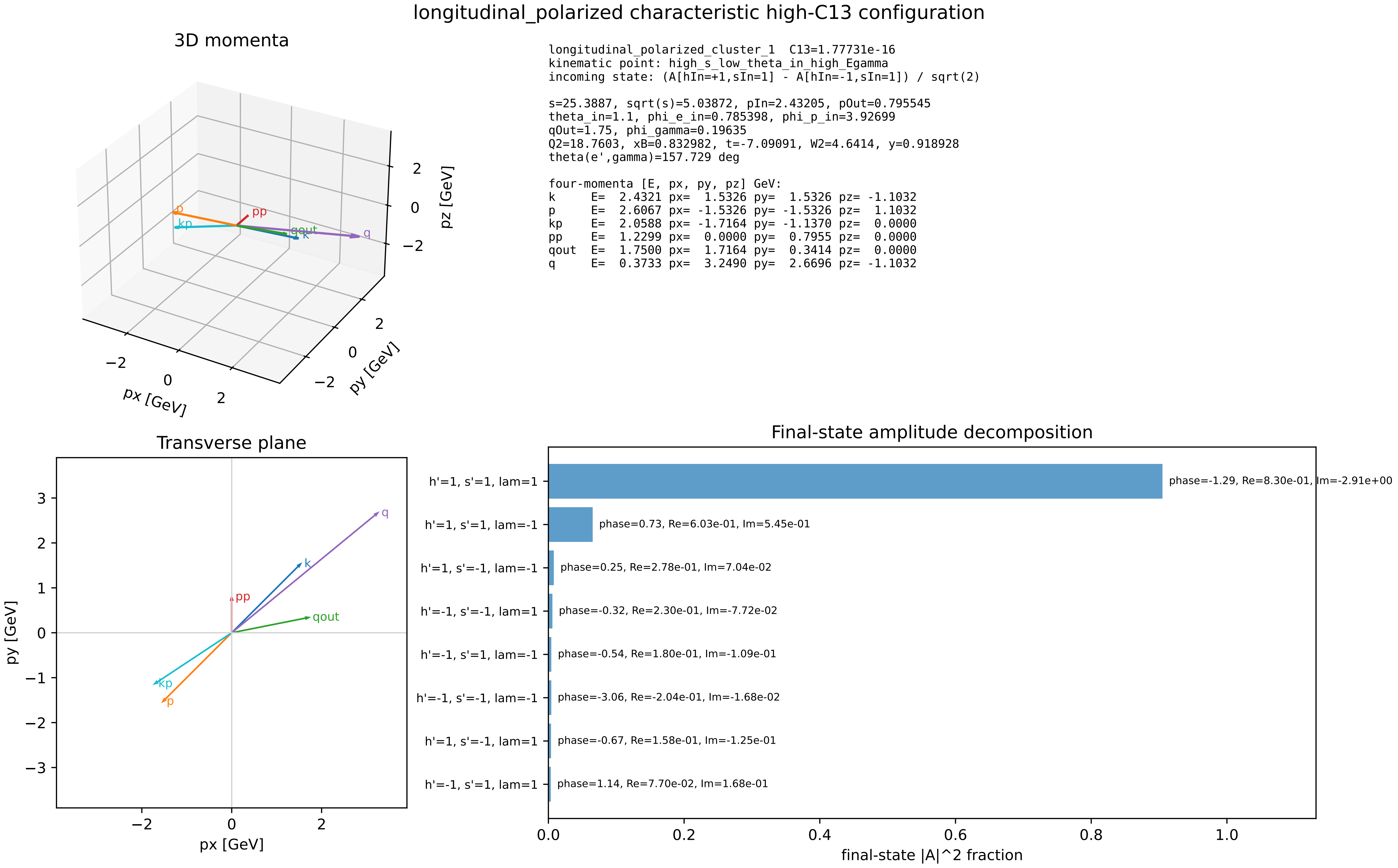
longitudinal_polarized characteristic high-C13 configuration



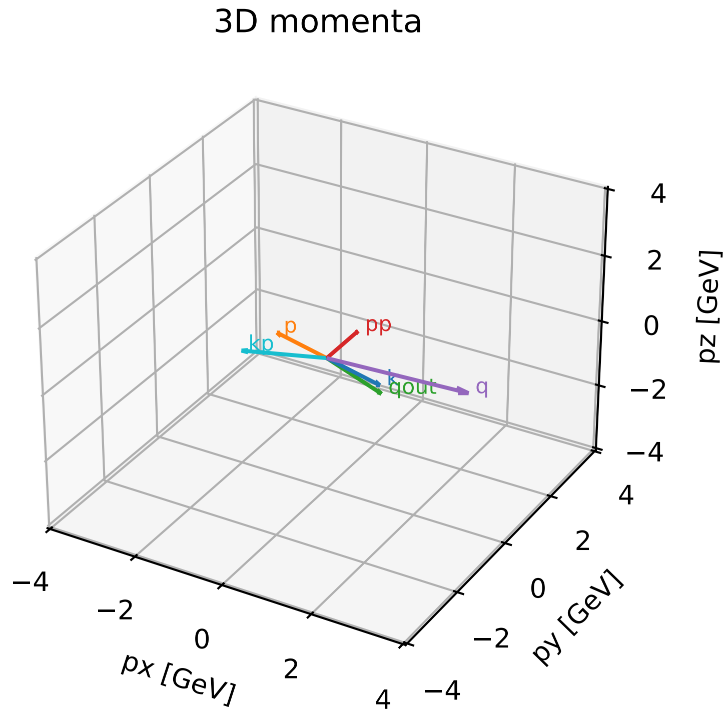

```
longitudinal_polarized_cluster_0  C13=1.96034e-16
kinematic point: mid_s_low_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] - A[hIn=-1,sIn=1]) / sqrt(2)

s=21.5158, sqrt(s)=4.63852, pIn=2.22442, pOut=1.07878
theta_in=1.1, phi_e_in=3.53429, phi_p_in=0.392699
qOut=1.75, phi_gamma=3.73064
Q2=11.659, xB=0.621479, t=-3.50566, W2=7.98096, y=0.909098
theta(e',gamma)=142.063 deg

four-momenta [E, px, py, pz] GeV:
k     E=  2.2244 px= -1.8315 py= -0.7586 pz= -1.0090
p     E=  2.4141 px=  1.8315 py=  0.7586 pz=  1.0090
kp    E=  1.4590 px=  1.4551 py= -0.1065 pz=  0.0000
pp    E=  1.4296 px=  0.0000 py=  1.0788 pz=  0.0000
qout  E=  1.7500 px= -1.4551 py= -0.9722 pz=  0.0000
q     E=  0.7655 px= -3.2866 py= -0.6521 pz= -1.0090
```



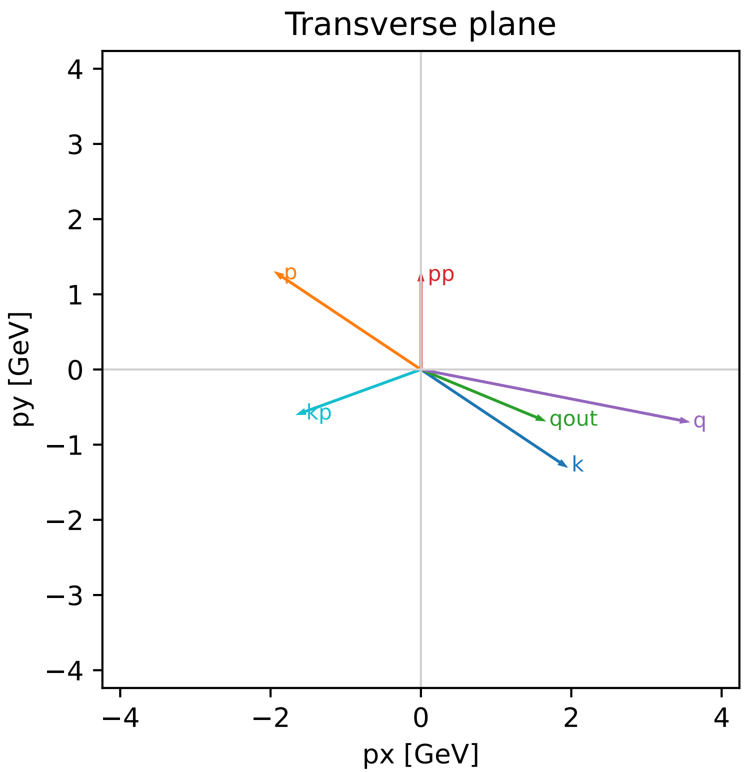


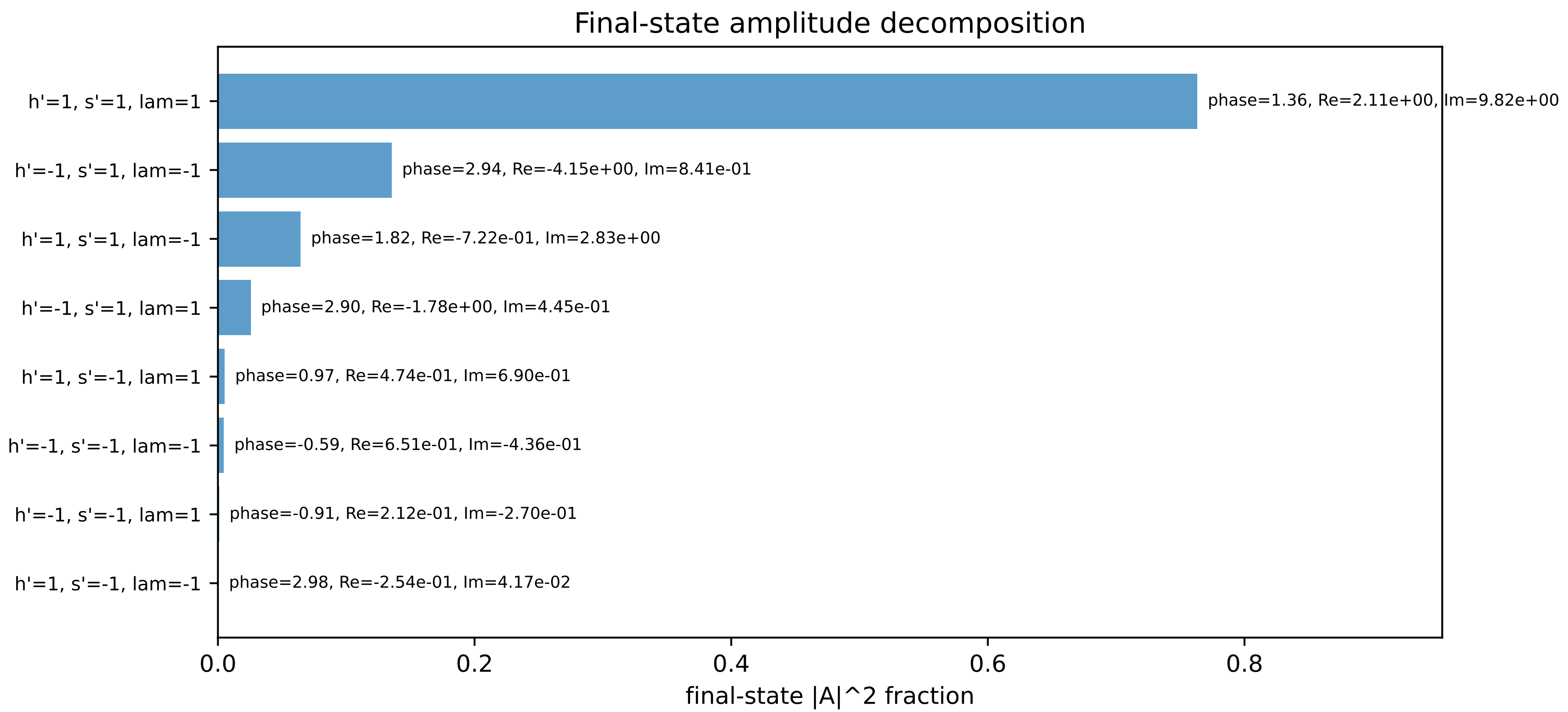

longitudinal_polarized characteristic high-C13 configuration
3D momenta
longitudinal_polarized_cluster_1 C13=1.77731e-16
kinematic point: high_s_low_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] - A[hIn=-1,sIn=1]) / sqrt(2)
s=25.3887, sqrt(s)=5.03872, pIn=2.43205, pOut=0.795545
theta_in=1.1, phi_e_in=0.785398, phi_p_in=3.92699
qOut=1.75, phi_gamma=0.19635
Q2=18.7603, xB=0.832982, t=-7.09091, W2=4.6414, y=0.918928
theta(e',gamma)=157.729 deg
four-momenta [E, px, py, pz] GeV:
k   E= 2.4321 px= 1.5326 py= 1.5326 pz= -1.1032
p   E= 2.6067 px= -1.5326 py= -1.5326 pz= 1.1032
kp   E= 2.0588 px= -1.7164 py= -1.1370 pz= 0.0000
pp   E= 1.2299 px= 0.0000 py= 0.7955 pz= 0.0000
qout E= 1.7500 px= 1.7164 py= 0.3414 pz= 0.0000
q   E= 0.3733 px= 3.2490 py= 2.6696 pz= -1.1032
px [GeV]
py [GeV]
pz [GeV]
Transverse plane
Final-state amplitude decomposition
h'=1, s'=1, lam=1
phase=-1.29, Re=8.30e-01, Im=-2.91e+00
h'=1, s'=1, lam=-1
phase=0.73, Re=6.03e-01, Im=5.45e-01
h'=1, s'=-1, lam=-1
phase=0.25, Re=2.78e-01, Im=7.04e-02
h'=-1, s'=-1, lam=1
phase=-0.32, Re=2.30e-01, Im=-7.72e-02
h'=-1, s'=1, lam=-1
phase=-0.54, Re=1.80e-01, Im=-1.09e-01
h'=-1, s'=-1, lam=-1
phase=-3.06, Re=-2.04e-01, Im=-1.68e-02
h'=1, s'=-1, lam=1
phase=-0.67, Re=1.58e-01, Im=-1.25e-01
h'=-1, s'=1, lam=1
phase=1.14, Re=7.70e-02, Im=1.68e-01
final-state |A|^2 fraction

longitudinal_polarized characteristic high-C13 configuration

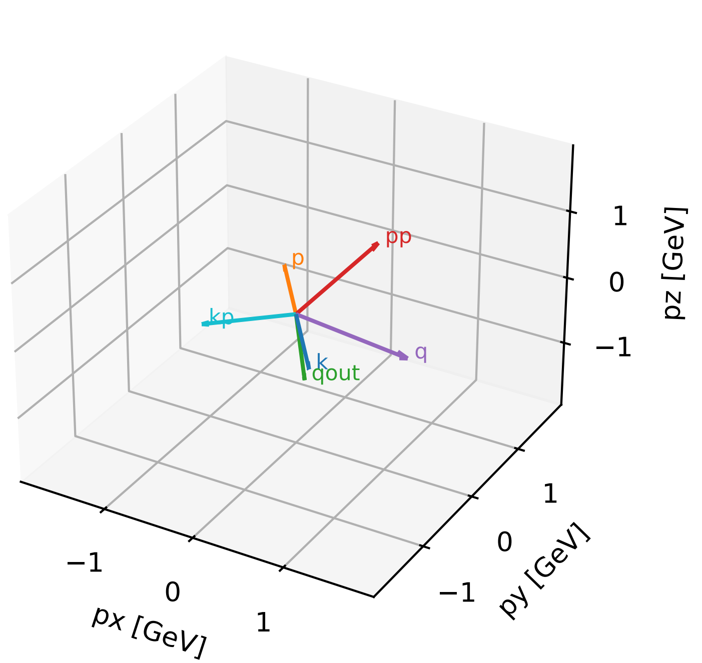



```
longitudinal_polarized_cluster_2  C13=1.76049e-16
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] - A[hIn=-1,sIn=1]) / sqrt(2)

s=25.3887, sqrt(s)=5.03872, pIn=2.43205, pOut=1.25714
theta_in=1.9, phi_e_in=5.69414, phi_p_in=2.55254
qOut=1.75, phi_gamma=5.89049
Q2=13.0527, xB=0.645333, t=-3.20272, W2=8.05346, y=0.825267
theta(e',gamma)=137.532 deg

four-momenta [E, px, py, pz] GeV:
k     E=  2.4321 px=  1.9136 py= -1.2786 pz=  0.7863
p     E=  2.6067 px= -1.9136 py=  1.2786 pz= -0.7863
kp    E=  1.7202 px= -1.6168 py= -0.5874 pz=  0.0000
pp    E=  1.5685 px=  0.0000 py=  1.2571 pz=  0.0000
qout  E=  1.7500 px=  1.6168 py= -0.6697 pz=  0.0000
q     E=  0.7118 px=  3.5304 py= -0.6912 pz=  0.7863
```

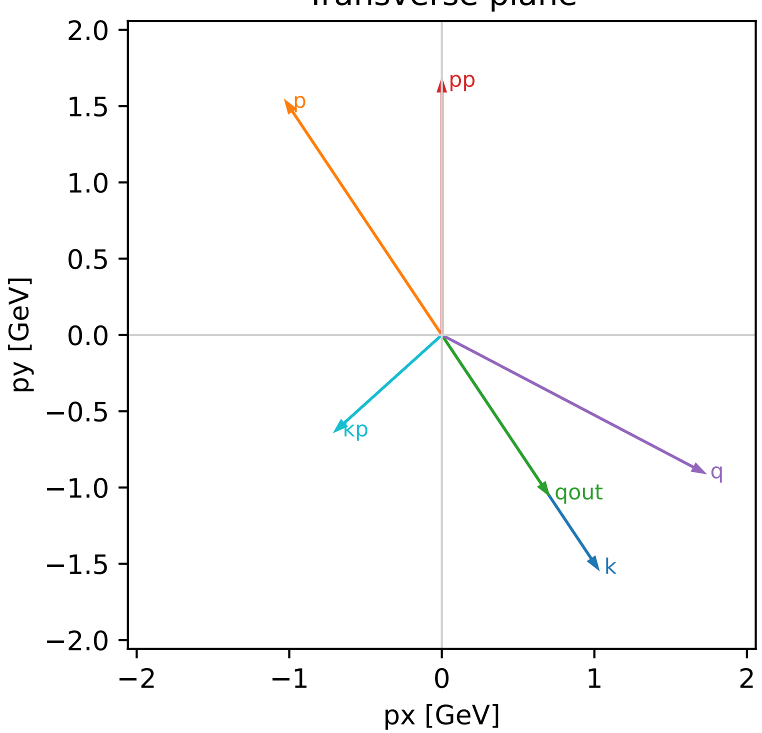


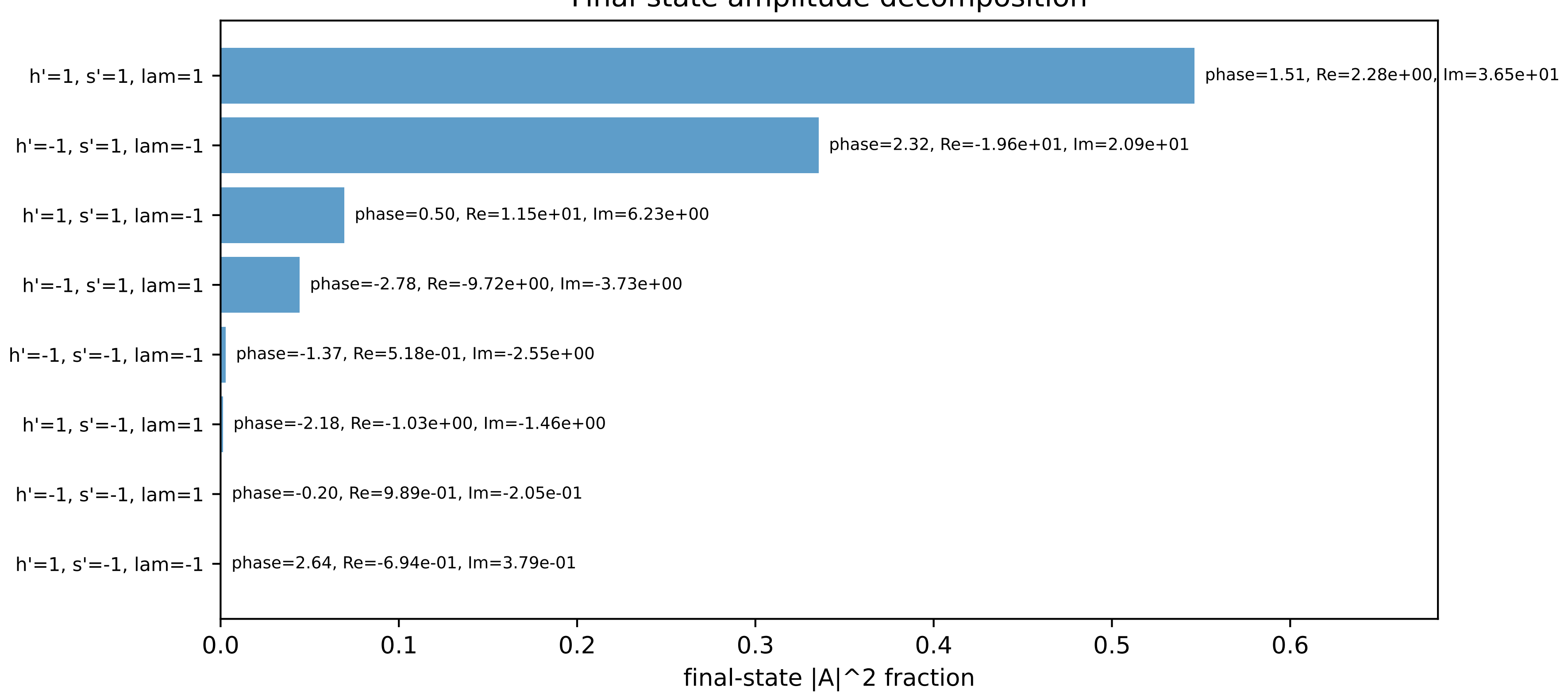


longitudinal_polarized characteristic high-C13 configuration

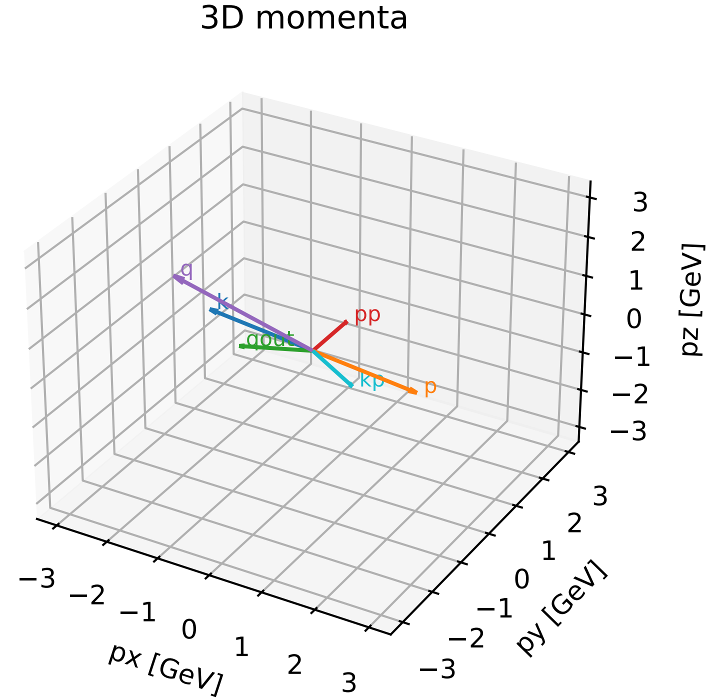



```
longitudinal_polarized_cluster_3  C13=1.62899e-16
kinematic point: low_s_high_theta_in_mid_Egamma
incoming state: (A[hIn=+1,sIn=1] - A[hIn=-1,sIn=1]) / sqrt(2)

s=16.7824, sqrt(s)=4.09663, pIn=1.94093, pOut=1.66555
theta_in=1.9, phi_e_in=5.30144, phi_p_in=2.15984
qOut=1.25, phi_gamma=5.30144
Q2=3.13457, xB=0.275554, t=-1.39451, W2=9.12079, y=0.715328
theta(e',gamma)=81.7082 deg

four-momenta [E, px, py, pz] GeV:
k     E=  1.9409 px=  1.0204 py= -1.5272 pz=  0.6275
p     E=  2.1557 px= -1.0204 py=  1.5272 pz= -0.6275
kp    E=  0.9351 px= -0.6945 py= -0.6262 pz=  0.0000
pp    E=  1.9115 px=  0.0000 py=  1.6656 pz=  0.0000
qout  E=  1.2500 px=  0.6945 py= -1.0393 pz=  0.0000
q     E=  1.0058 px=  1.7149 py= -0.9009 pz=  0.6275
```

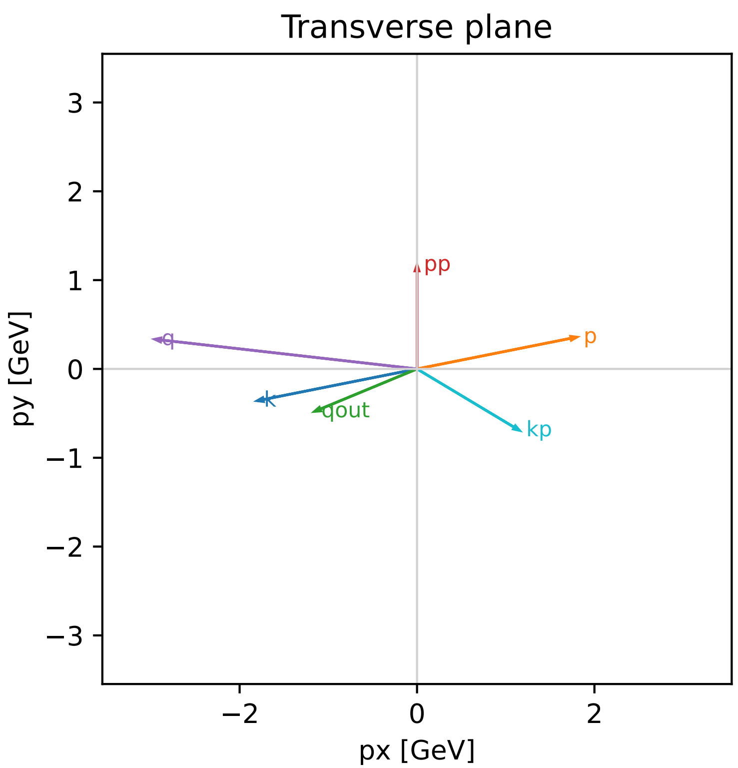


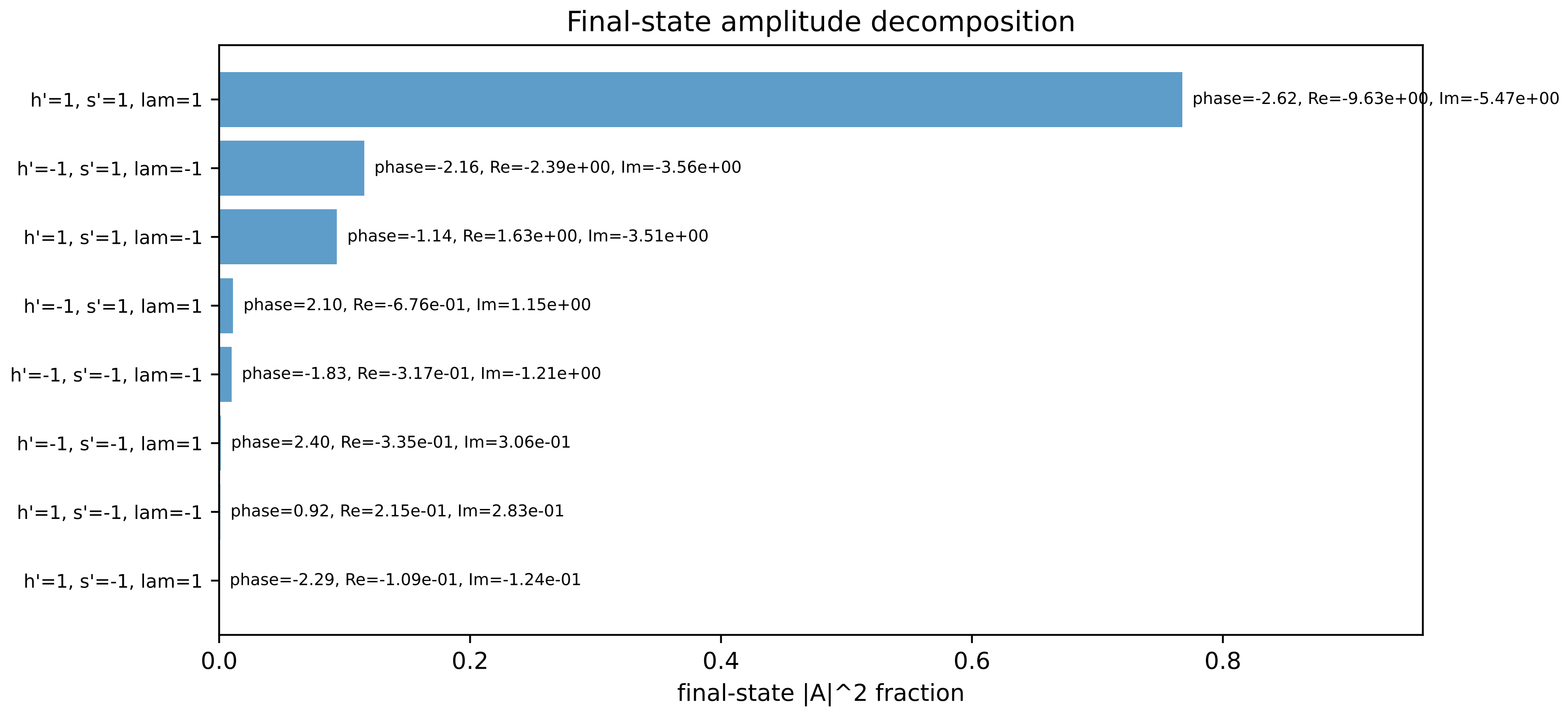


longitudinal_polarized characteristic high-C13 configuration

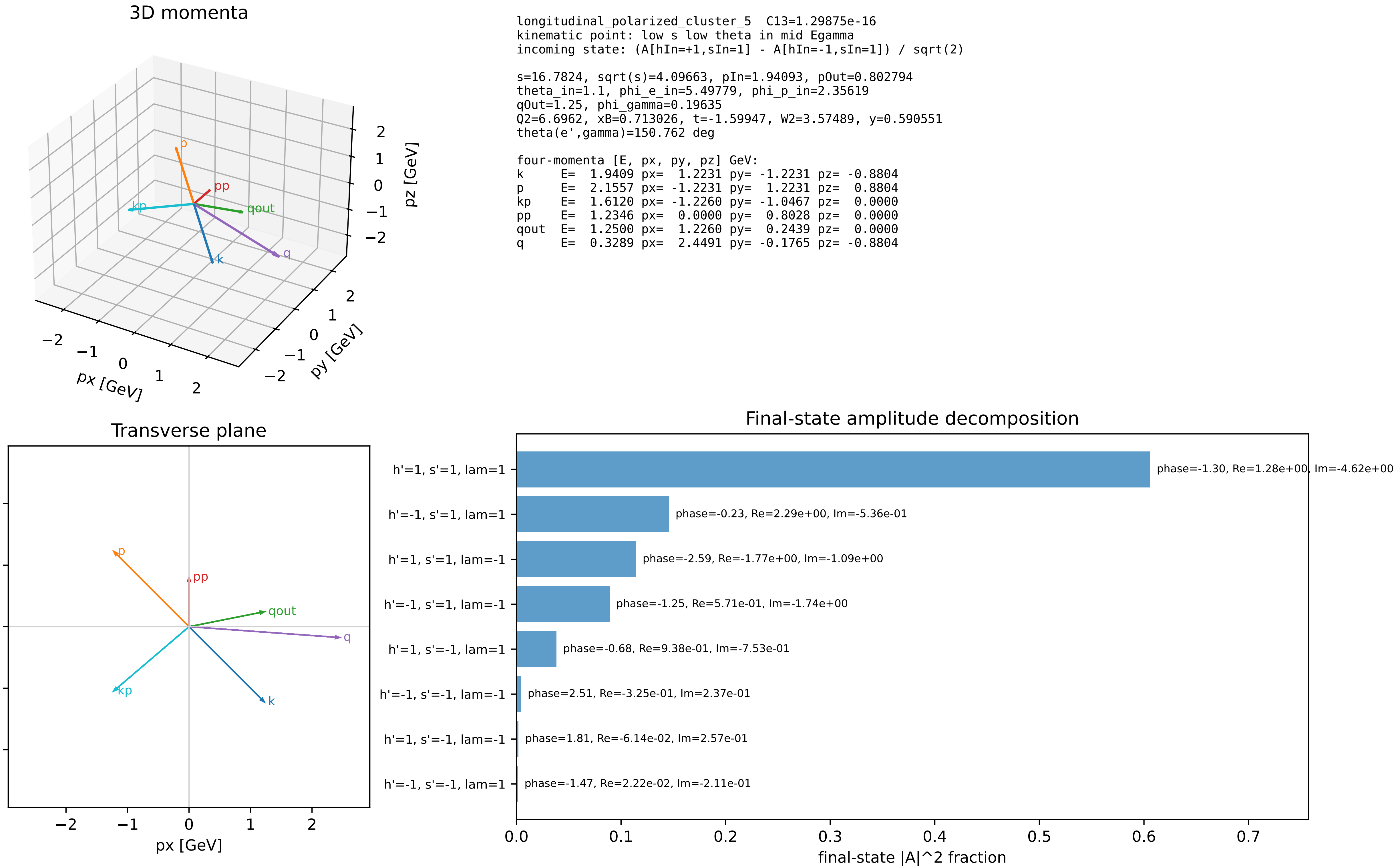



```
longitudinal_polarized_cluster_4  C13=1.5085e-16
kinematic point: low_s_high_theta_in_mid_Egamma
incoming state: (A[hIn=+1,sIn=1] - A[hIn=-1,sIn=1]) / sqrt(2)

s=16.7824, sqrt(s)=4.09663, pIn=1.94093, pOut=1.17071
theta_in=1.9, phi_e_in=3.33794, phi_p_in=0.19635
qOut=1.25, phi_gamma=3.53429
Q2=8.89142, xB=0.646094, t=-3.86902, W2=5.75022, y=0.865385
theta(e',gamma)=126.556 deg

four-momenta [E, px, py, pz] GeV:
k     E=  1.9409 px= -1.8014 py= -0.3583 pz=  0.6275
p     E=  2.1557 px=  1.8014 py=  0.3583 pz= -0.6275
kp    E=  1.3465 px=  1.1548 py= -0.6924 pz=  0.0000
pp    E=  1.5001 px=  0.0000 py=  1.1707 pz=  0.0000
qout  E=  1.2500 px= -1.1548 py= -0.4784 pz=  0.0000
q     E=  0.5944 px= -2.9563 py=  0.3340 pz=  0.6275
```



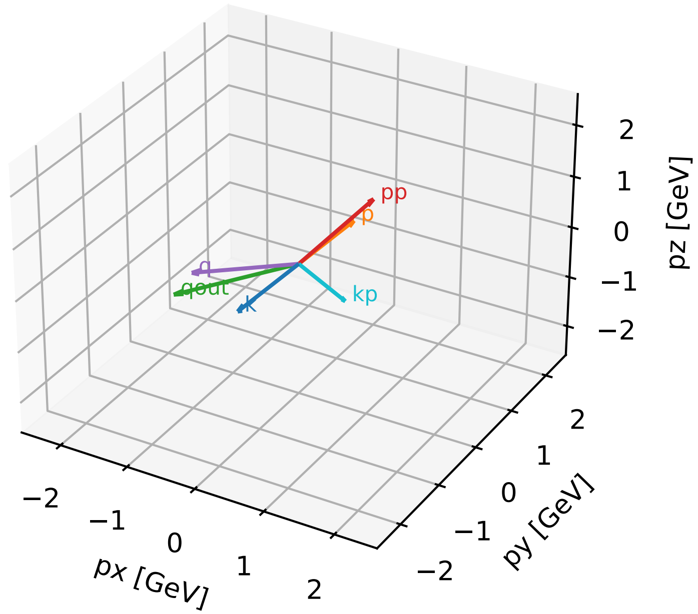


longitudinal_polarized characteristic high-C13 configuration



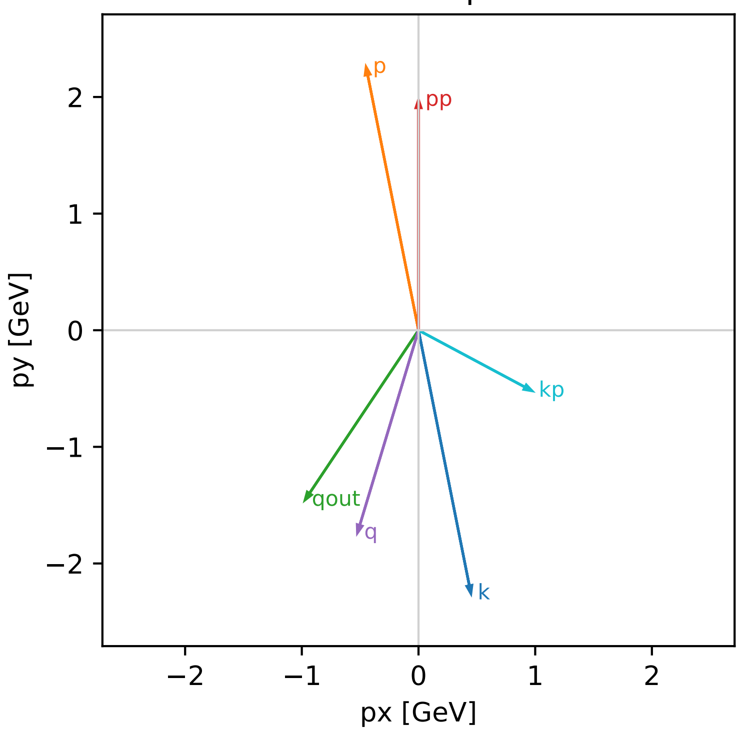
longitudinal_polarized characteristic high-C13 configuration

3D momenta

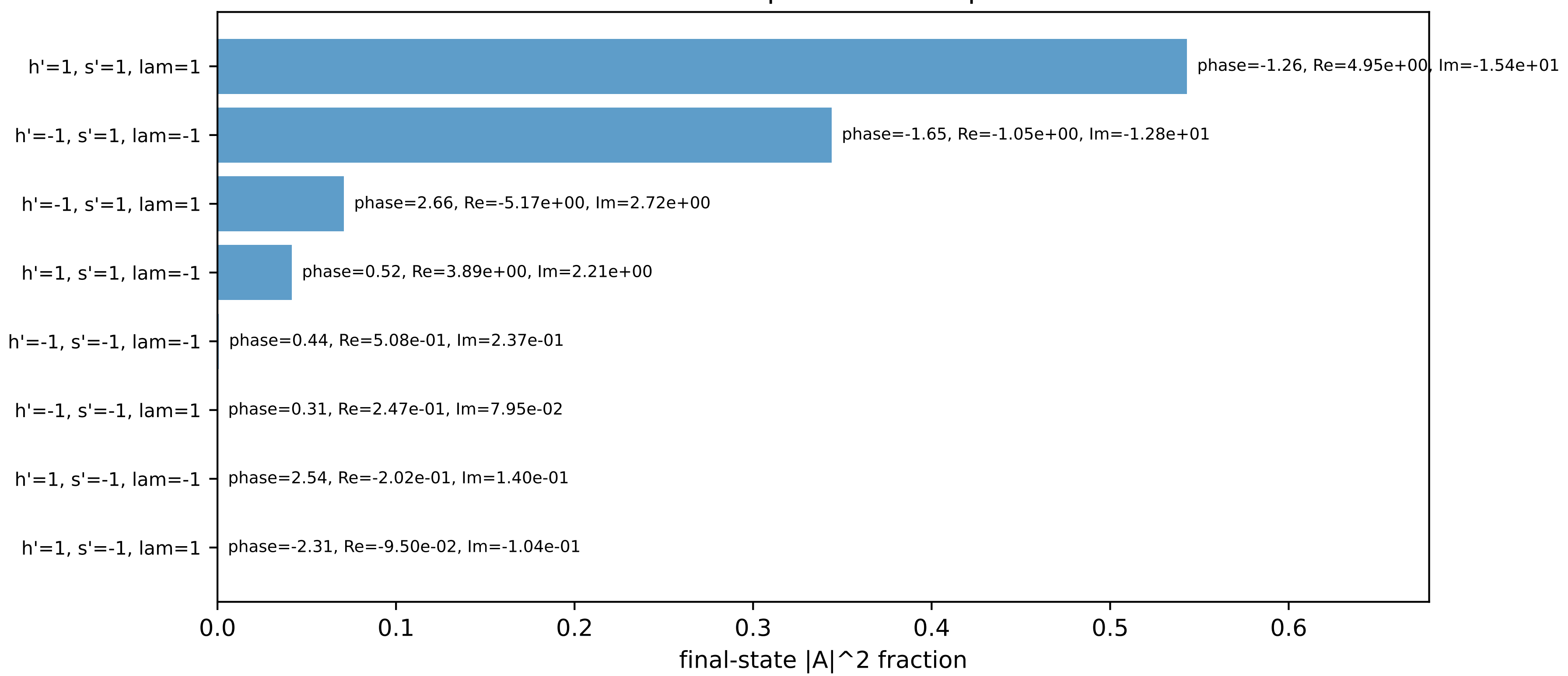


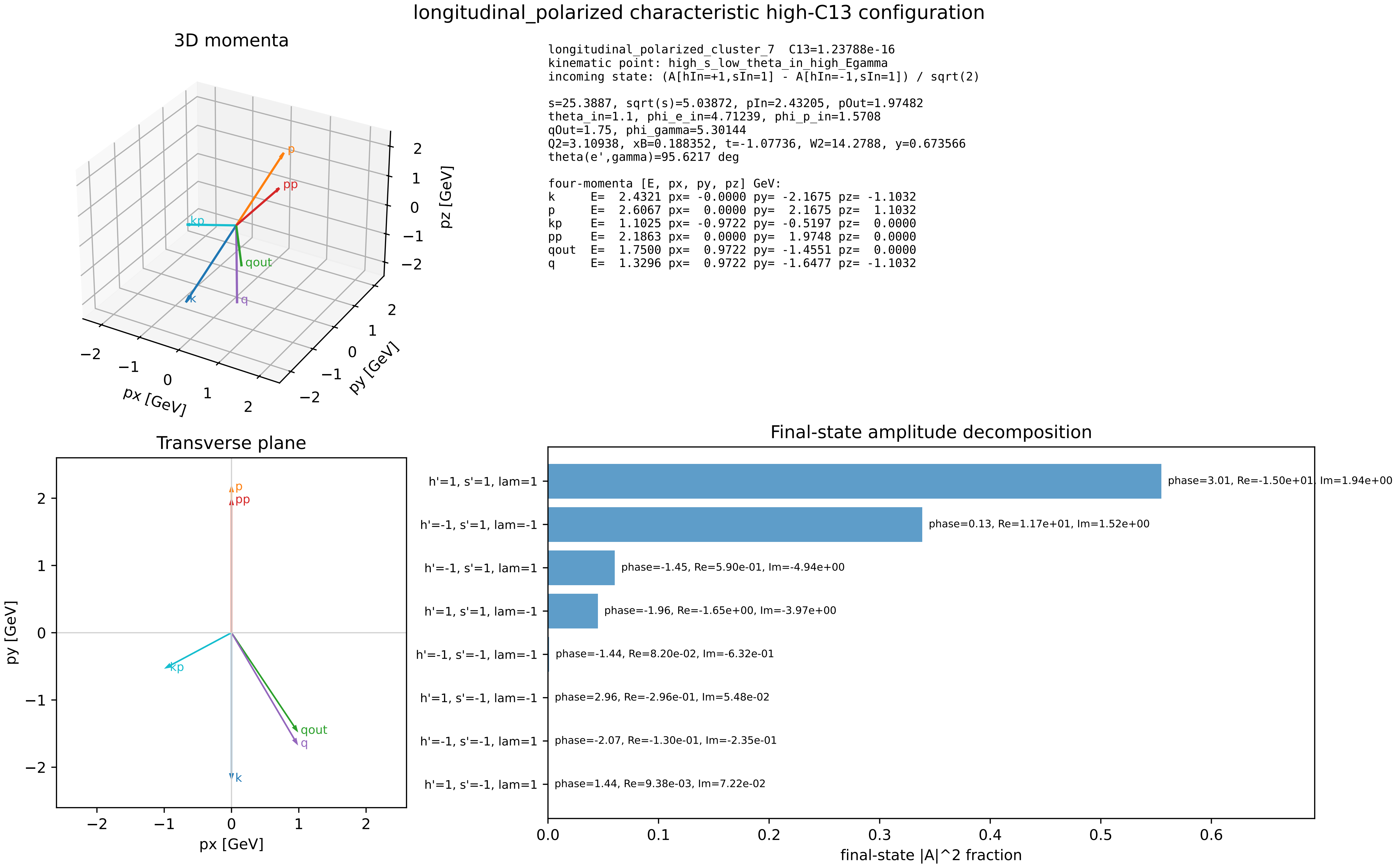

```
longitudinal_polarized_cluster_6  C13=1.24971e-16
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] - A[hIn=-1,sIn=1]) / sqrt(2)

s=25.3887, sqrt(s)=5.03872, pIn=2.43205, pOut=1.97482
theta_in=1.9, phi_e_in=4.90874, phi_p_in=1.76715
qOut=1.75, phi_gamma=4.12334
Q2=2.14301, xB=0.137885, t=-0.722809, W2=14.2788, y=0.634136
theta(e',gamma)=95.6217 deg

four-momenta [E, px, py, pz] GeV:
k     E=  2.4321 px=  0.4490 py= -2.2572 pz=  0.7863
p     E=  2.6067 px= -0.4490 py=  2.2572 pz= -0.7863
kp    E=  1.1025 px=  0.9722 py= -0.5197 pz=  0.0000
pp    E=  2.1863 px=  0.0000 py=  1.9748 pz=  0.0000
qout  E=  1.7500 px= -0.9722 py= -1.4551 pz=  0.0000
q     E=  1.3296 px= -0.5233 py= -1.7375 pz=  0.7863
```

Transverse plane

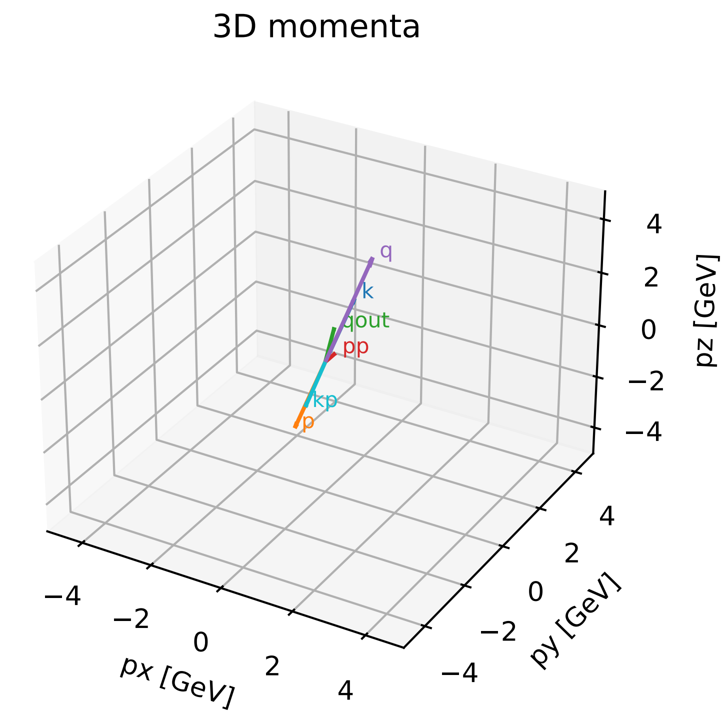


Final-state amplitude decomposition

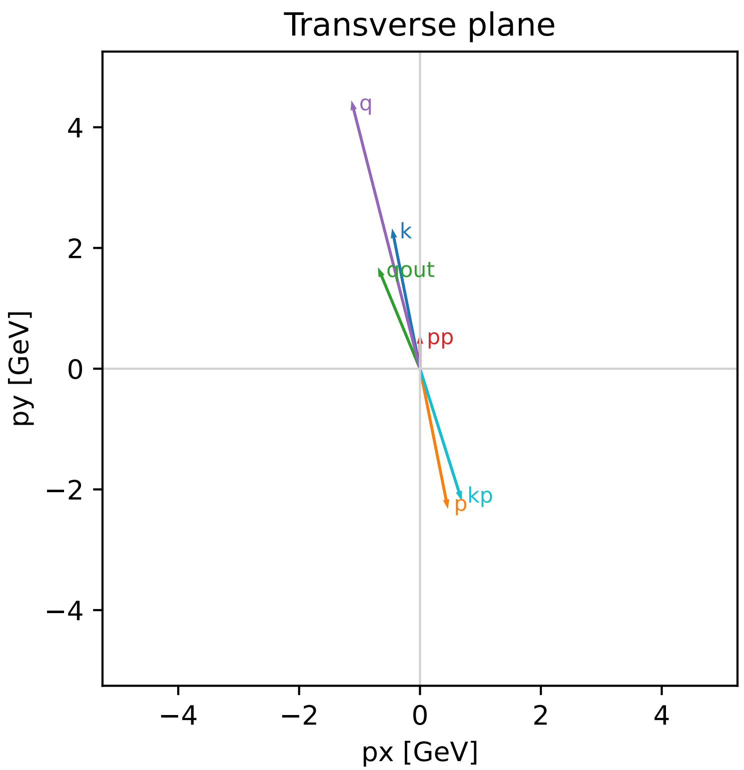


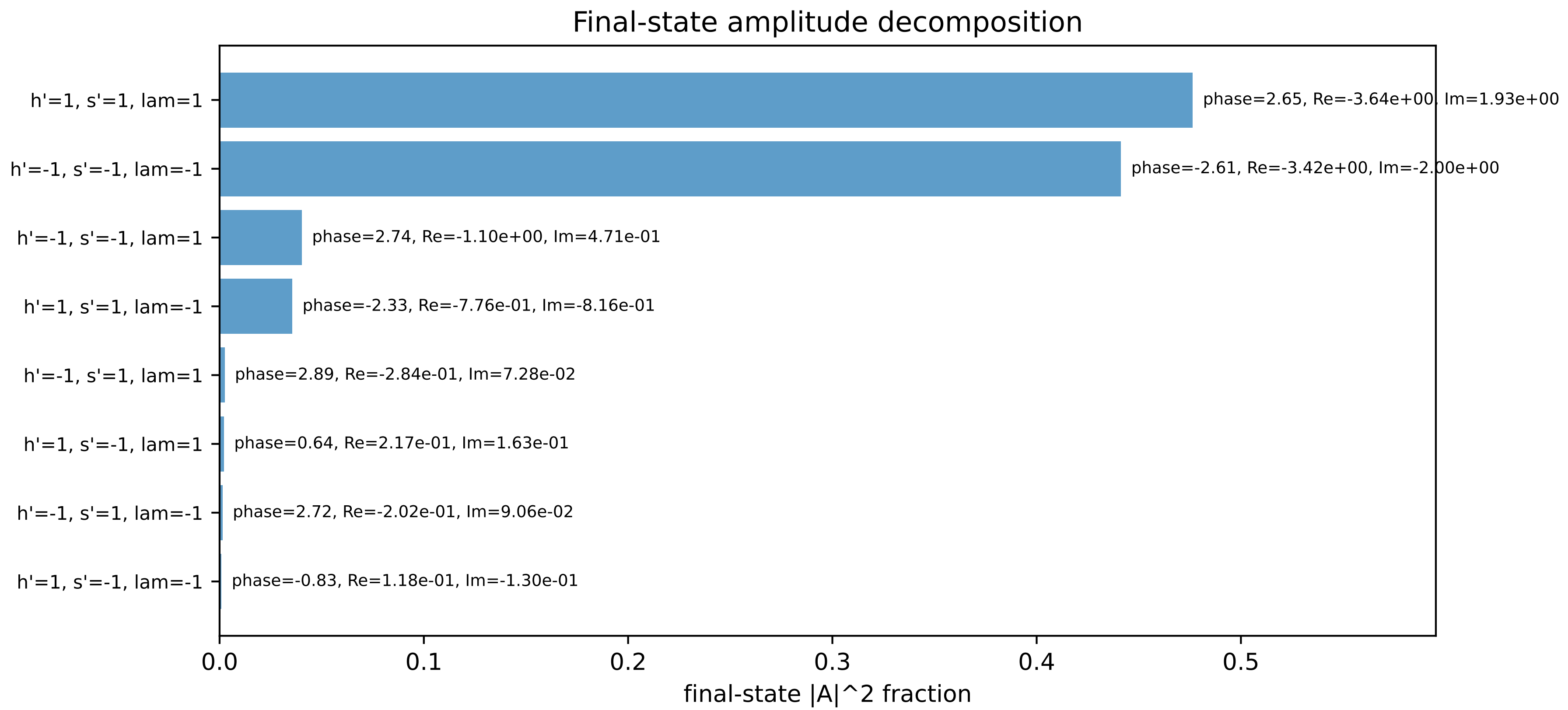

longitudinal_polarized characteristic high-C13 configuration
3D momenta
longitudinal_polarized_cluster_7 C13=1.23788e-16
kinematic point: high_s_low_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] - A[hIn=-1,sIn=1]) / sqrt(2)
s=25.3887, sqrt(s)=5.03872, pIn=2.43205, pOut=1.97482
theta_in=1.1, phi_e_in=4.71239, phi_p_in=1.5708
qOut=1.75, phi_gamma=5.30144
Q2=3.10938, xB=0.188352, t=-1.07736, W2=14.2788, y=0.673566
theta(e',gamma)=95.6217 deg
four-momenta [E, px, py, pz] GeV:
k   E= 2.4321 px= -0.0000 py= -2.1675 pz= -1.1032
p   E= 2.6067 px= 0.0000 py= 2.1675 pz= 1.1032
kp   E= 1.1025 px= -0.9722 py= -0.5197 pz= 0.0000
pp   E= 2.1863 px= 0.0000 py= 1.9748 pz= 0.0000
qout E= 1.7500 px= 0.9722 py= -1.4551 pz= 0.0000
q   E= 1.3296 px= 0.9722 py= -1.6477 pz= -1.1032
px [GeV]
py [GeV]
pz [GeV]
p
pp
kp
qout
k
q
Transverse plane
Final-state amplitude decomposition
h'=1, s'=1, lam=1
phase=3.01, Re=-1.50e+01, Im=1.94e+00
h'=-1, s'=1, lam=-1
phase=0.13, Re=1.17e+01, Im=1.52e+00
h'=-1, s'=1, lam=1
phase=-1.45, Re=5.90e-01, Im=-4.94e+00
h'=1, s'=1, lam=-1
phase=-1.96, Re=-1.65e+00, Im=-3.97e+00
h'=-1, s'=-1, lam=-1
phase=-1.44, Re=8.20e-02, Im=-6.32e-01
h'=1, s'=-1, lam=-1
phase=2.96, Re=-2.96e-01, Im=5.48e-02
h'=-1, s'=-1, lam=1
phase=-2.07, Re=-1.30e-01, Im=-2.35e-01
h'=1, s'=-1, lam=1
phase=1.44, Re=9.38e-03, Im=7.22e-02
final-state |A|^2 fraction

unpolarized characteristic high-C13 configuration

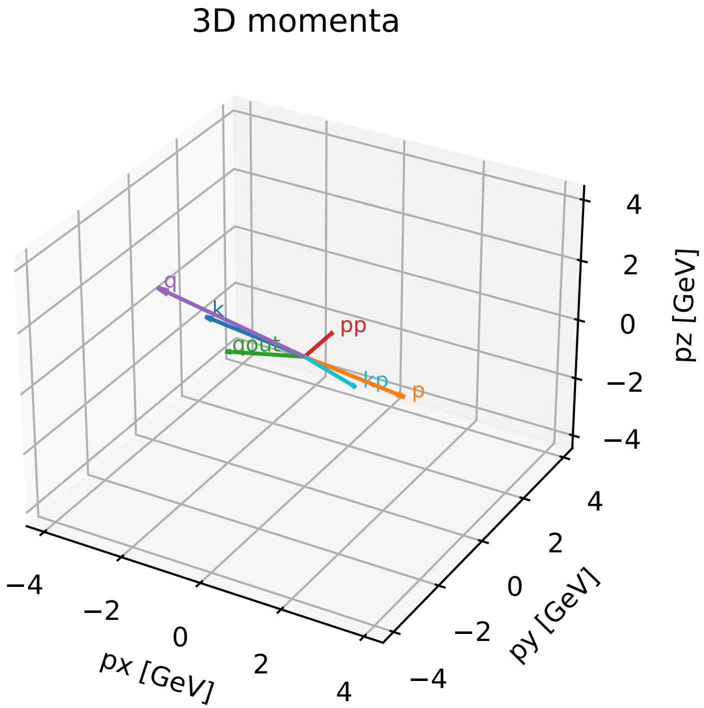



```
unpolarized_cluster_0  C13=4.60614e-16
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: coherent equal sum over all (hIn,sIn) rows, divided by sqrt(4)

s=25.3887, sqrt(s)=5.03872, pIn=2.43205, pOut=0.503914
theta_in=1.9, phi_e_in=1.76715, phi_p_in=4.90874
qOut=1.75, phi_gamma=1.9635
Q2=20.9926, xB=0.909168, t=-6.06631, W2=2.97716, y=0.942106
theta(e',gamma)=175.026 deg

four-momenta [E, px, py, pz] GeV:
k     E=  2.4321 px= -0.4490 py=  2.2572 pz=  0.7863
p     E=  2.6067 px=  0.4490 py= -2.2572 pz= -0.7863
kp    E=  2.2239 px=  0.6697 py= -2.1207 pz=  0.0000
pp    E=  1.0648 px=  0.0000 py=  0.5039 pz=  0.0000
qout  E=  1.7500 px= -0.6697 py=  1.6168 pz=  0.0000
q     E=  0.2081 px= -1.1187 py=  4.3779 pz=  0.7863
```

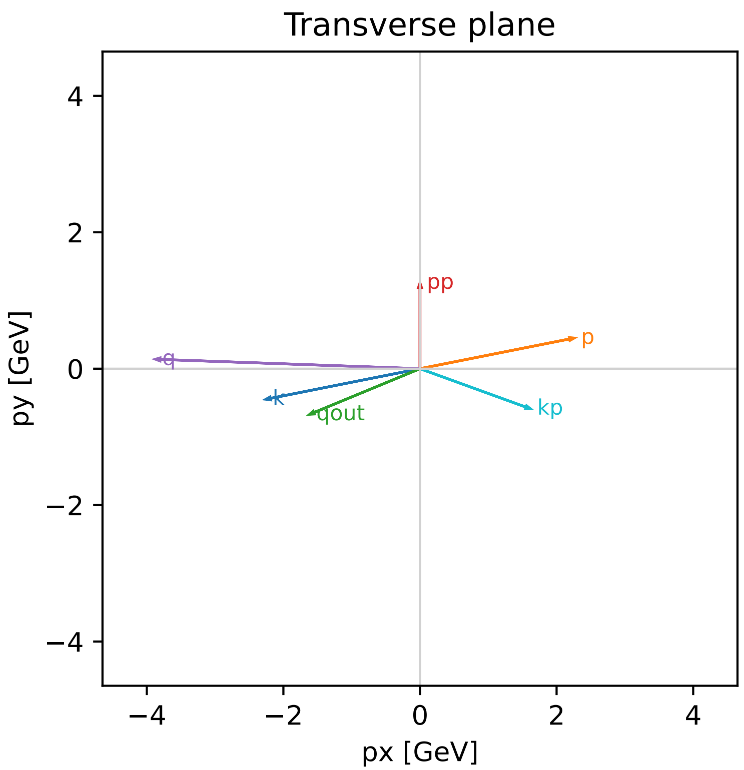


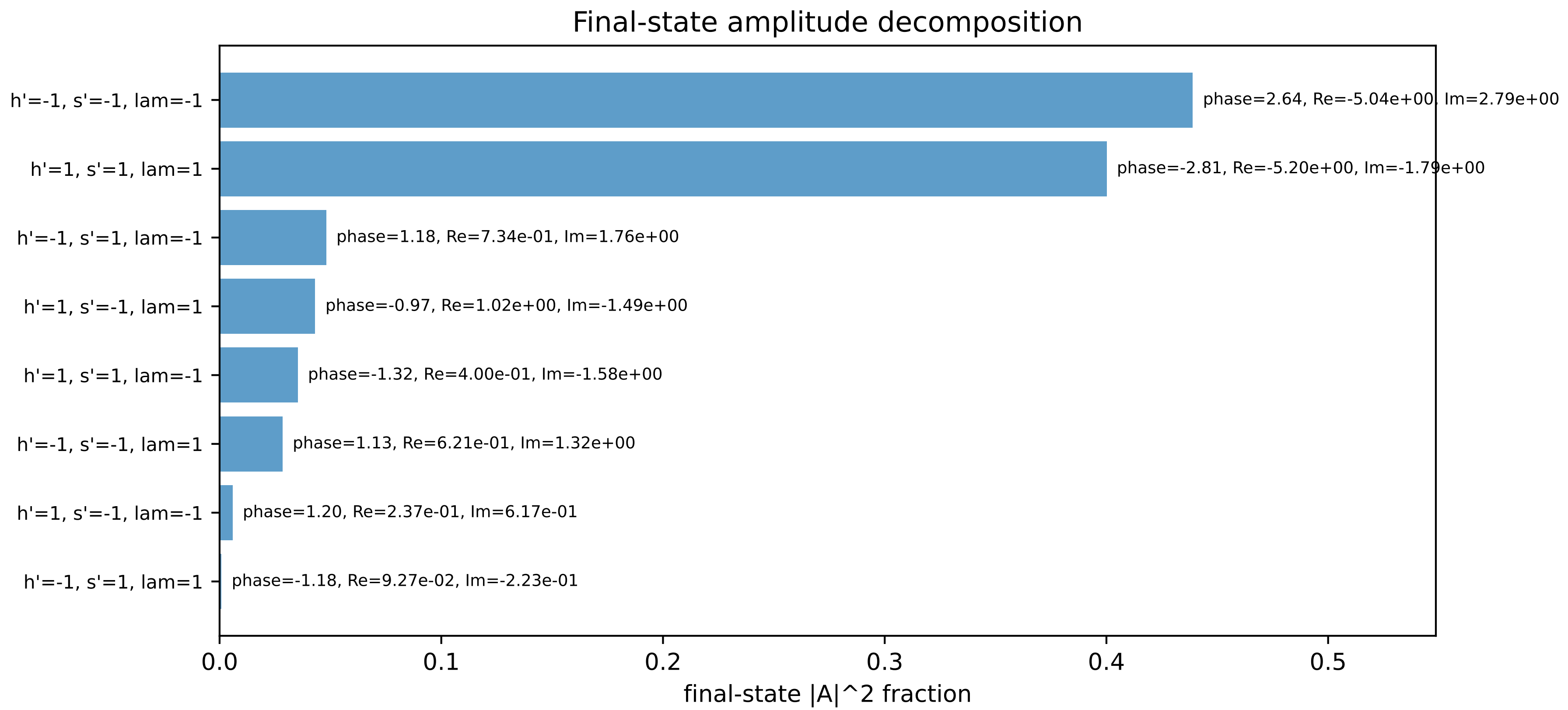


unpolarized characteristic high-C13 configuration

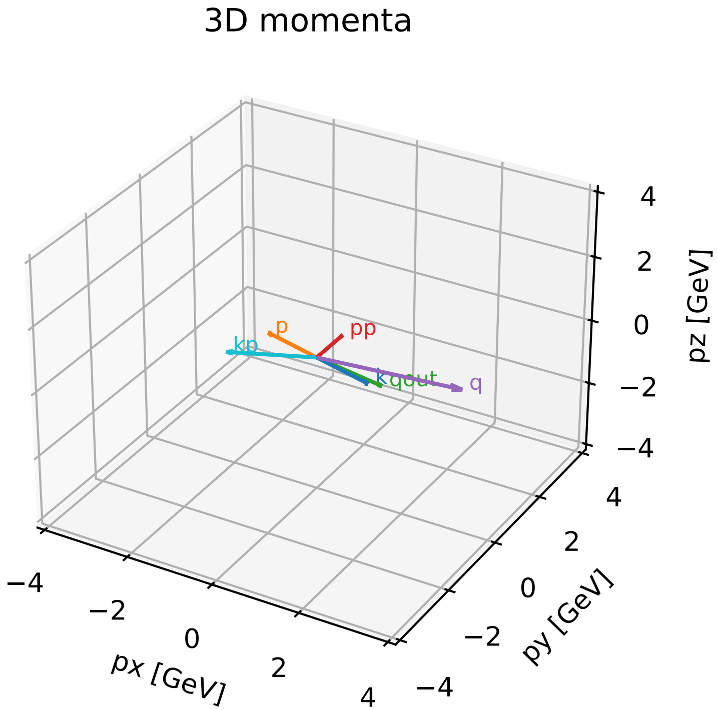



```
unpolarized_cluster_1  C13=4.33938e-16
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: coherent equal sum over all (hIn,sIn) rows, divided by sqrt(4)

s=25.3887, sqrt(s)=5.03872, pIn=2.43205, pOut=1.25714
theta_in=1.9, phi_e_in=3.33794, phi_p_in=0.19635
qOut=1.75, phi_gamma=3.53429
Q2=15.1387, xB=0.67849, t=-5.28863, W2=8.05346, y=0.910376
theta(e',gamma)=137.532 deg

four-momenta [E, px, py, pz] GeV:
k     E=  2.4321 px= -2.2572 py= -0.4490 pz=  0.7863
p     E=  2.6067 px=  2.2572 py=  0.4490 pz= -0.7863
kp    E=  1.7202 px=  1.6168 py= -0.5874 pz=  0.0000
pp    E=  1.5685 px=  0.0000 py=  1.2571 pz=  0.0000
qout  E=  1.7500 px= -1.6168 py= -0.6697 pz=  0.0000
q     E=  0.7118 px= -3.8740 py=  0.1385 pz=  0.7863
```

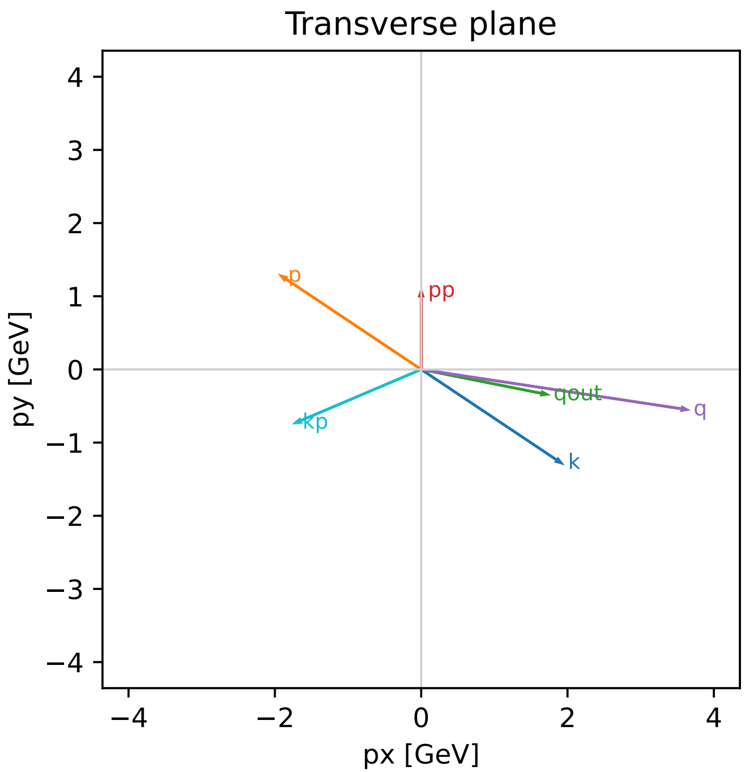


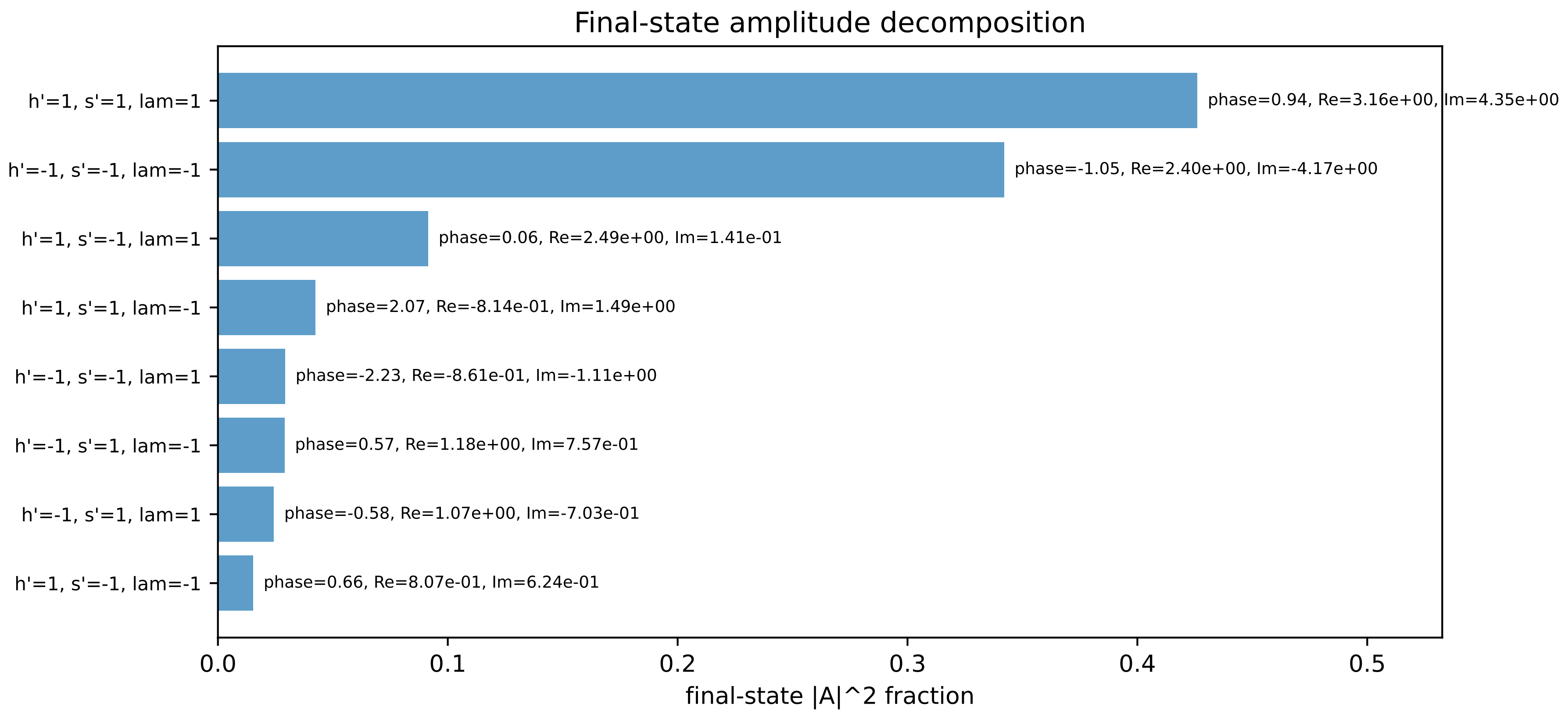


unpolarized characteristic high-C13 configuration

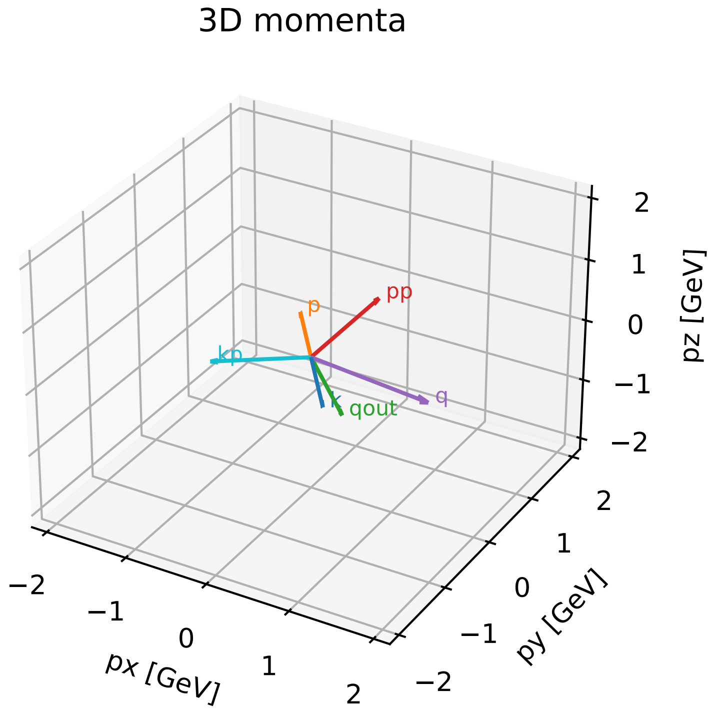



```
unpolarized_cluster_2  C13=4.17371e-16
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: coherent equal sum over all (hIn,sIn) rows, divided by sqrt(4)

s=25.3887, sqrt(s)=5.03872, pIn=2.43205, pOut=1.07102
theta_in=1.9, phi_e_in=5.69414, phi_p_in=2.55254
qOut=1.75, phi_gamma=6.08684
Q2=13.7747, xB=0.706794, t=-2.92371, W2=6.59412, y=0.795181
theta(e',gamma)=145.72 deg

four-momenta [E, px, py, pz] GeV:
k     E=  2.4321 px=  1.9136 py= -1.2786 pz=  0.7863
p     E=  2.6067 px= -1.9136 py=  1.2786 pz= -0.7863
kp    E=  1.8650 px= -1.7164 py= -0.7296 pz=  0.0000
pp    E=  1.4237 px=  0.0000 py=  1.0710 pz=  0.0000
qout  E=  1.7500 px=  1.7164 py= -0.3414 pz=  0.0000
q     E=  0.5670 px=  3.6300 py= -0.5490 pz=  0.7863
```

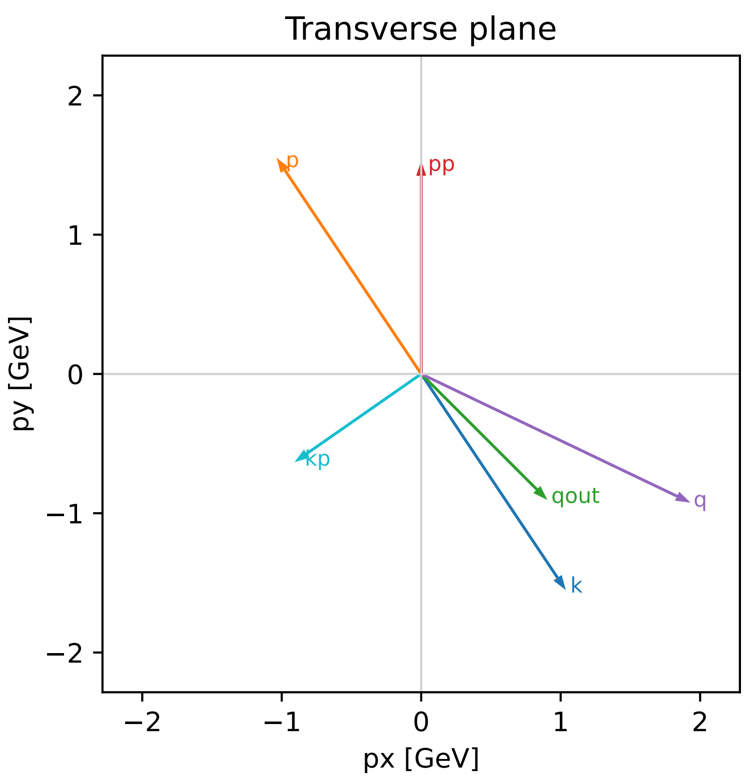


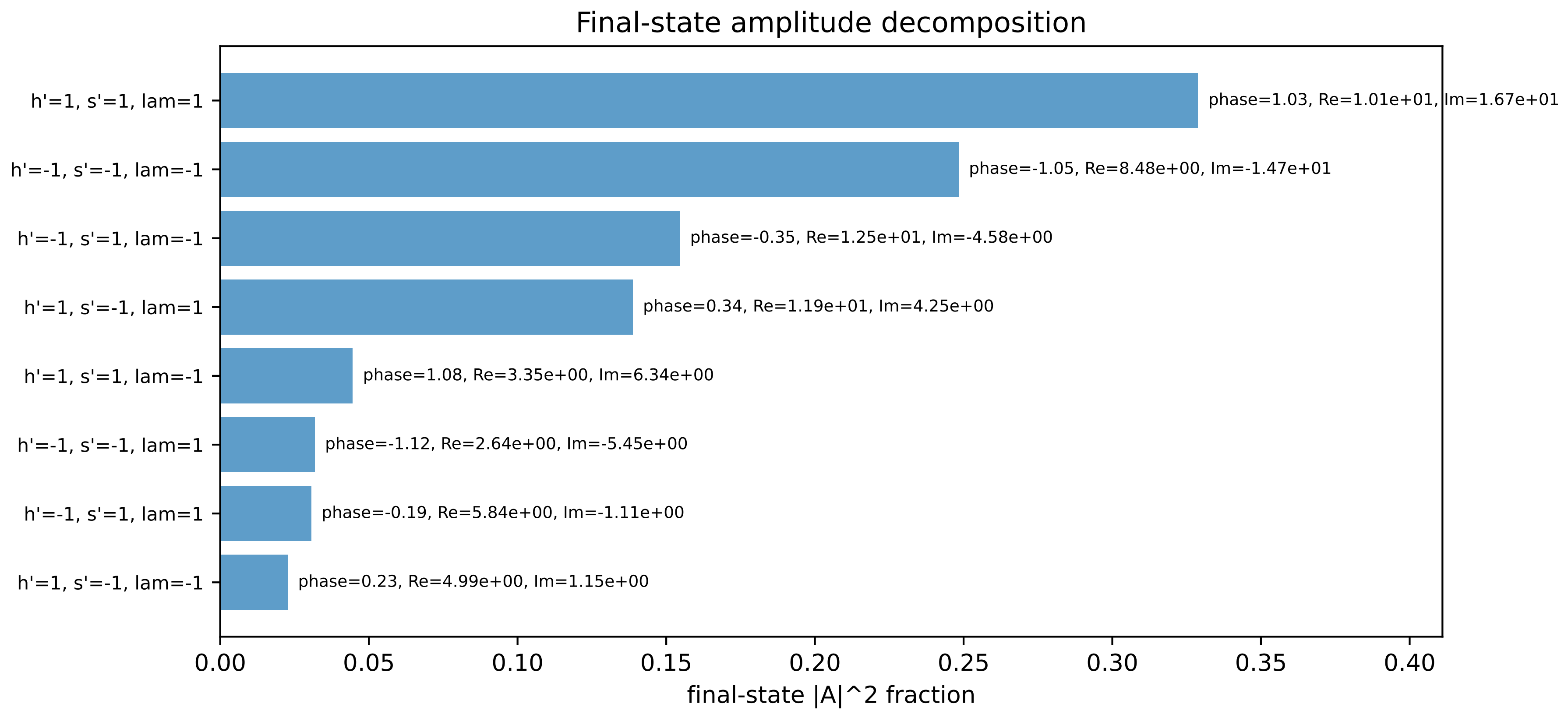


unpolarized characteristic high-C13 configuration

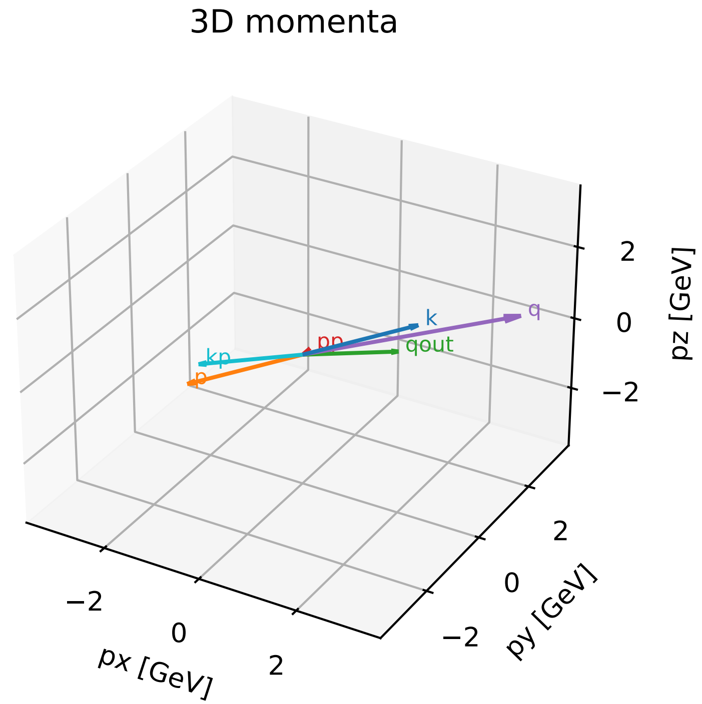



```
unpolarized_cluster_3  C13=3.41782e-16
kinematic point: low_s_high_theta_in_mid_Egamma
incoming state: coherent equal sum over all (hIn,sIn) rows, divided by sqrt(4)

s=16.7824, sqrt(s)=4.09663, pIn=1.94093, pOut=1.50004
theta_in=1.9, phi_e_in=5.30144, phi_p_in=2.15984
qOut=1.25, phi_gamma=5.49779
Q2=4.10442, xB=0.367152, t=-1.28631, W2=7.95451, y=0.702976
theta(e',gamma)=100.119 deg

four-momenta [E, px, py, pz] GeV:
k     E=  1.9409 px=  1.0204 py= -1.5272 pz=  0.6275
p     E=  2.1557 px= -1.0204 py=  1.5272 pz= -0.6275
kp    E=  1.0775 px= -0.8839 py= -0.6162 pz=  0.0000
pp    E=  1.7692 px=  0.0000 py=  1.5000 pz=  0.0000
qout  E=  1.2500 px=  0.8839 py= -0.8839 pz=  0.0000
q     E=  0.8635 px=  1.9043 py= -0.9110 pz=  0.6275
```

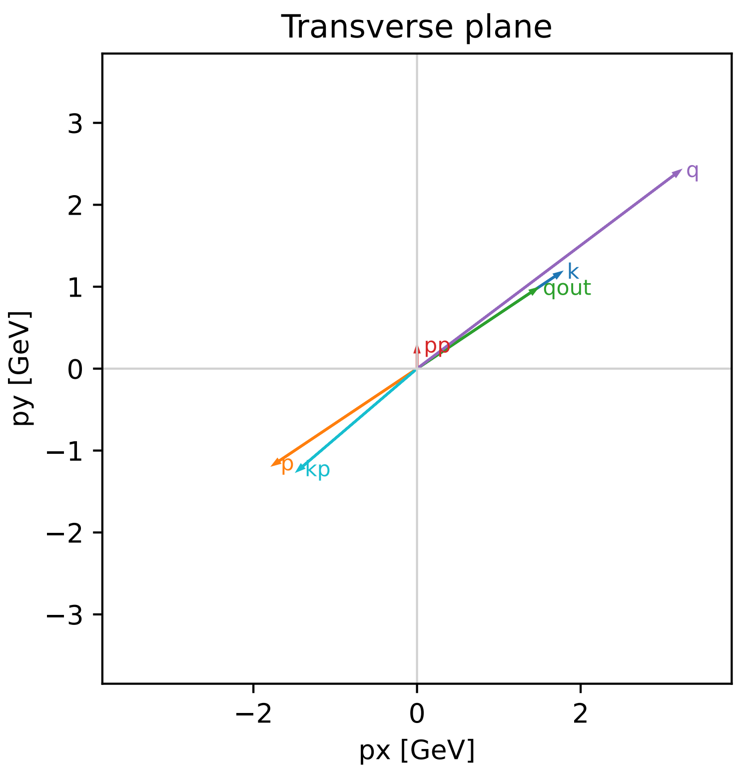


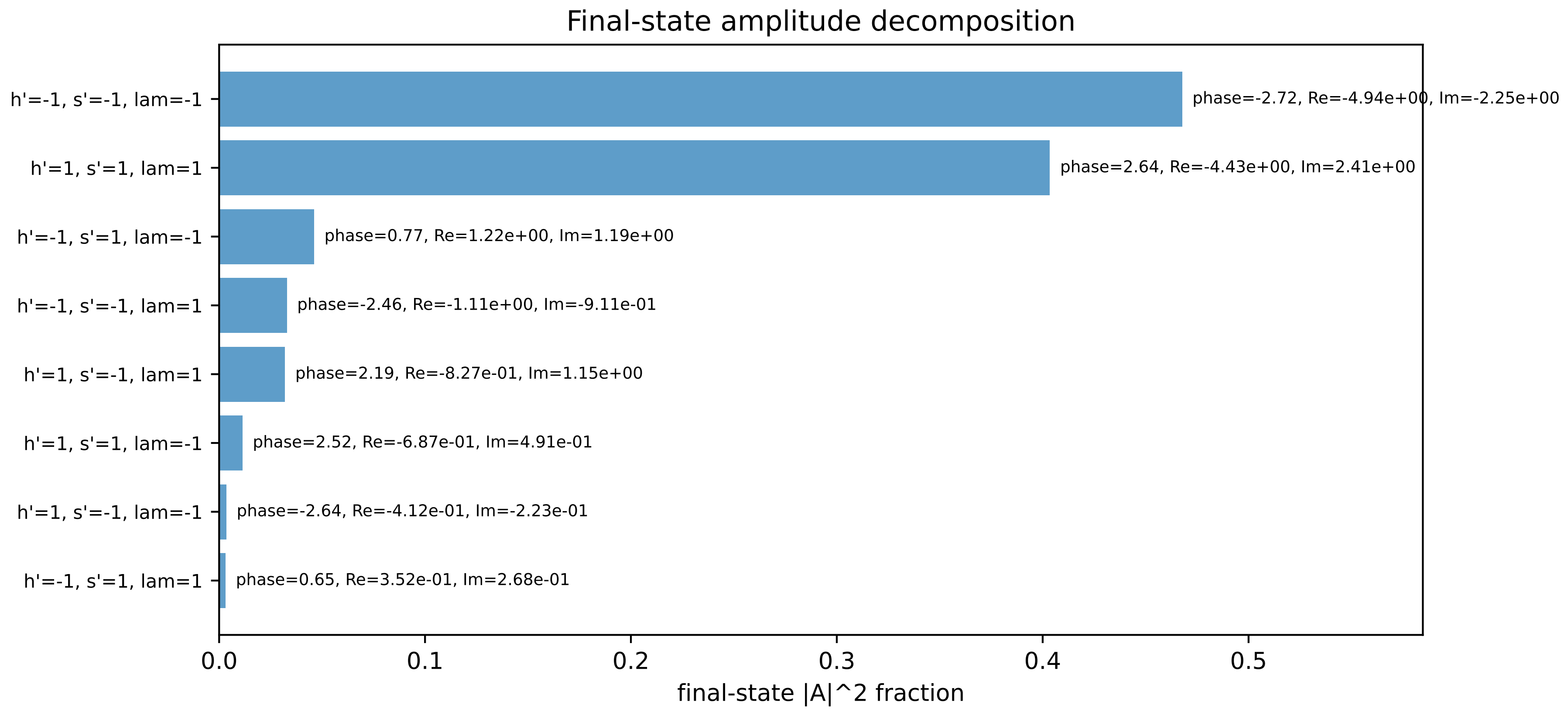


unpolarized characteristic high-C13 configuration

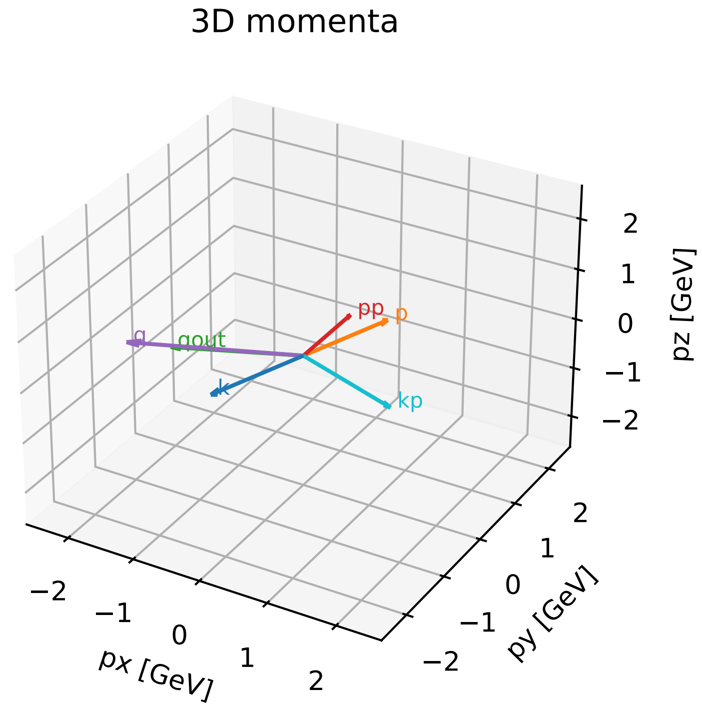



```
unpolarized_cluster_4  C13=3.31224e-16
kinematic point: mid_s_high_theta_in_high_Egamma
incoming state: coherent equal sum over all (hIn,sIn) rows, divided by sqrt(4)

s=21.5158, sqrt(s)=4.63852, pIn=2.22442, pOut=0.269147
theta_in=1.9, phi_e_in=0.589049, phi_p_in=3.73064
qOut=1.75, phi_gamma=0.589049
Q2=16.506, xB=0.850907, t=-3.58142, W2=3.77197, y=0.940015
theta(e',gamma)=173.281 deg

four-momenta [E, px, py, pz] GeV:
k     E=  2.2244 px=  1.7502 py=  1.1695 pz=  0.7191
p     E=  2.4141 px= -1.7502 py= -1.1695 pz= -0.7191
kp    E=  1.9127 px= -1.4551 py= -1.2414 pz=  0.0000
pp    E=  0.9759 px=  0.0000 py=  0.2691 pz=  0.0000
qout  E=  1.7500 px=  1.4551 py=  0.9722 pz=  0.0000
q     E=  0.3118 px=  3.2053 py=  2.4109 pz=  0.7191
```

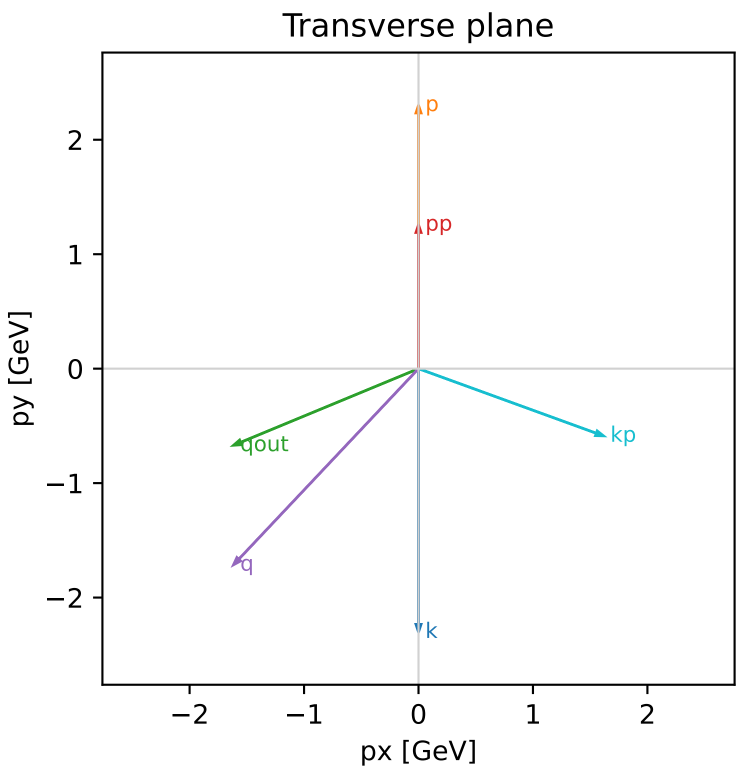


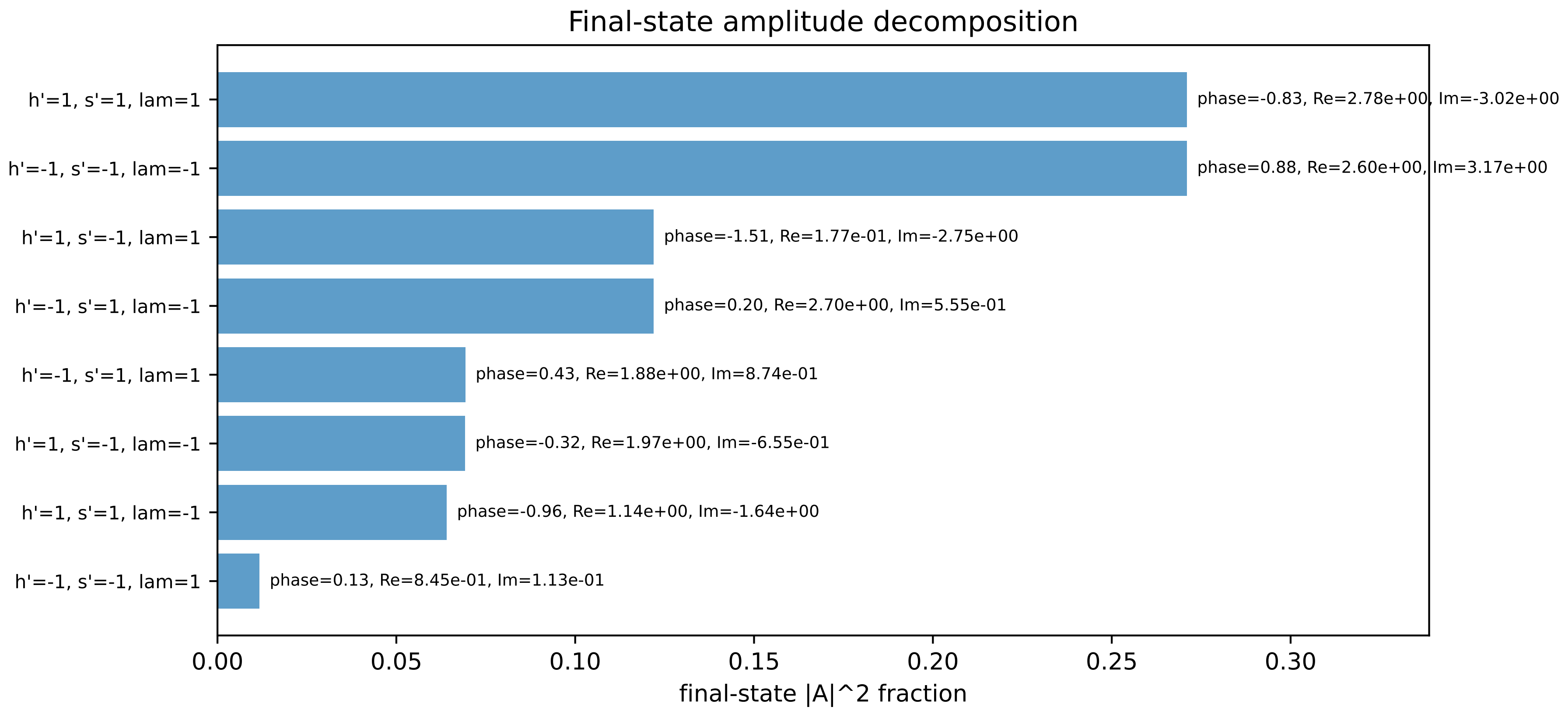


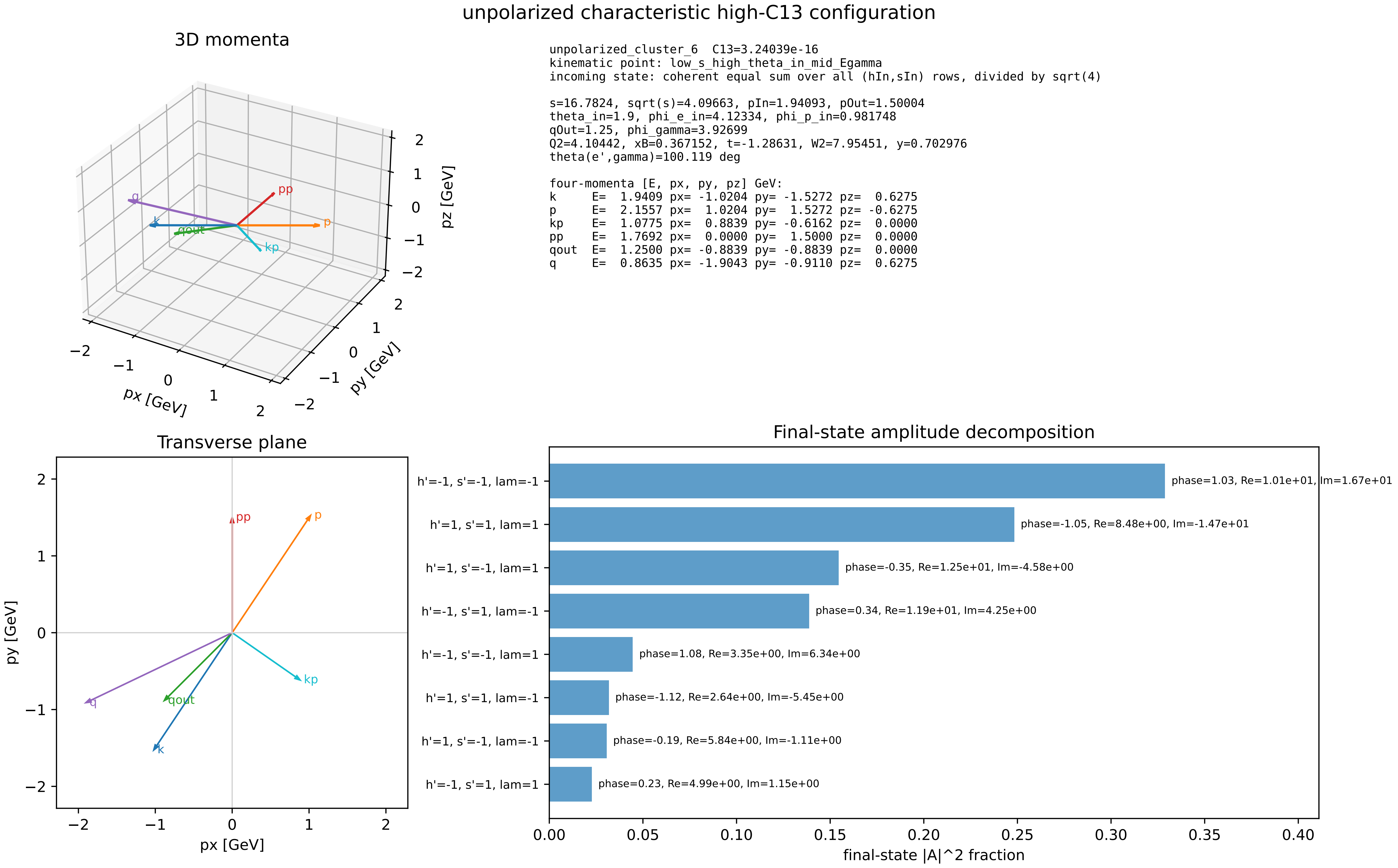
unpolarized characteristic high-C13 configuration



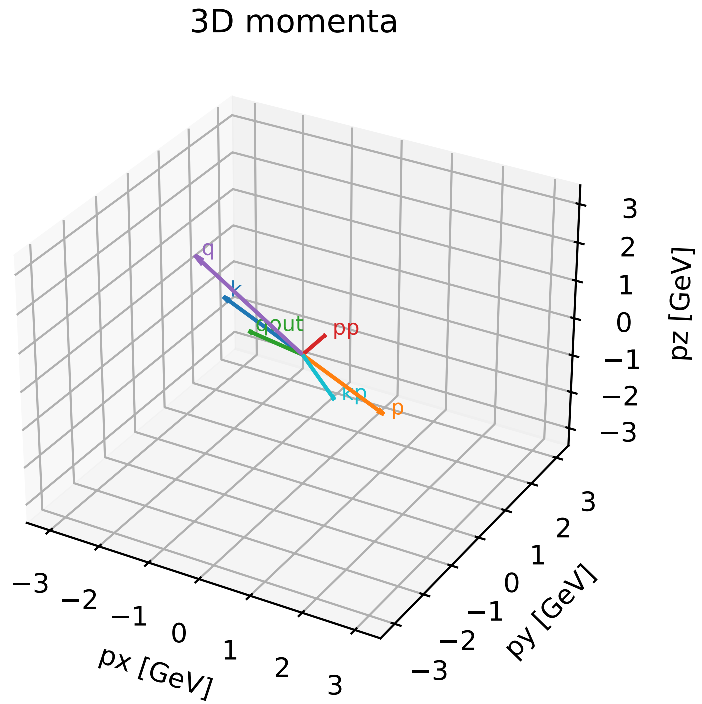

```
unpolarized_cluster_5  C13=3.30496e-16
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: coherent equal sum over all (hIn,sIn) rows, divided by sqrt(4)

s=25.3887, sqrt(s)=5.03872, pIn=2.43205, pOut=1.25714
theta_in=1.9, phi_e_in=4.71239, phi_p_in=1.5708
qOut=1.75, phi_gamma=3.53429
Q2=5.6633, xB=0.441173, t=-0.631026, W2=8.05346, y=0.523766
theta(e',gamma)=137.532 deg

four-momenta [E, px, py, pz] GeV:
k     E=  2.4321 px= -0.0000 py= -2.3015 pz=  0.7863
p     E=  2.6067 px=  0.0000 py=  2.3015 pz= -0.7863
kp    E=  1.7202 px=  1.6168 py= -0.5874 pz=  0.0000
pp    E=  1.5685 px=  0.0000 py=  1.2571 pz=  0.0000
qout  E=  1.7500 px= -1.6168 py= -0.6697 pz=  0.0000
q     E=  0.7118 px= -1.6168 py= -1.7140 pz=  0.7863
```






unpolarized characteristic high-C13 configuration
3D momenta
unpolarized_cluster_6 C13=3.24039e-16
kinematic point: low_s_high_theta_in_mid_Egamma
incoming state: coherent equal sum over all (hIn,sIn) rows, divided by sqrt(4)
s=16.7824, sqrt(s)=4.09663, pIn=1.94093, pOut=1.50004
theta_in=1.9, phi_e_in=4.12334, phi_p_in=0.981748
qOut=1.25, phi_gamma=3.92699
Q2=4.10442, xB=0.367152, t=-1.28631, W2=7.95451, y=0.702976
theta(e',gamma)=100.119 deg
four-momenta [E, px, py, pz] GeV:
k   E= 1.9409 px= -1.0204 py= -1.5272 pz= 0.6275
p   E= 2.1557 px= 1.0204 py= 1.5272 pz= -0.6275
kp   E= 1.0775 px= 0.8839 py= -0.6162 pz= 0.0000
pp   E= 1.7692 px= 0.0000 py= 1.5000 pz= 0.0000
qout E= 1.2500 px= -0.8839 py= -0.8839 pz= 0.0000
q   E= 0.8635 px= -1.9043 py= -0.9110 pz= 0.6275
px [GeV]
py [GeV]
pz [GeV]
Transverse plane
Final-state amplitude decomposition
h'=-1, s'=-1, lam=-1
phase=1.03, Re=1.01e+01, Im=1.67e+01
h'=1, s'=1, lam=1
phase=-1.05, Re=8.48e+00, Im=-1.47e+01
h'=1, s'=-1, lam=1
phase=-0.35, Re=1.25e+01, Im=-4.58e+00
h'=-1, s'=1, lam=-1
phase=0.34, Re=1.19e+01, Im=4.25e+00
h'=-1, s'=-1, lam=1
phase=1.08, Re=3.35e+00, Im=6.34e+00
h'=1, s'=1, lam=-1
phase=-1.12, Re=2.64e+00, Im=-5.45e+00
h'=1, s'=-1, lam=-1
phase=-0.19, Re=5.84e+00, Im=-1.11e+00
h'=-1, s'=1, lam=1
phase=0.23, Re=4.99e+00, Im=1.15e+00
final-state |A|^2 fraction

unpolarized characteristic high-C13 configuration




```
unpolarized_cluster_7  C13=2.8173e-16
kinematic point: low_s_high_theta_in_mid_Egamma
incoming state: coherent equal sum over all (hIn,sIn) rows, divided by sqrt(4)

s=16.7824, sqrt(s)=4.09663, pIn=1.94093, pOut=0.802794
theta_in=1.9, phi_e_in=2.94524, phi_p_in=6.08684
qOut=1.25, phi_gamma=2.94524
Q2=11.4246, xB=0.809128, t=-4.13863, W2=3.57489, y=0.887887
theta(e',gamma)=150.762 deg

four-momenta [E, px, py, pz] GeV:
k     E=  1.9409 px= -1.8014 py=  0.3583 pz=  0.6275
p     E=  2.1557 px=  1.8014 py= -0.3583 pz= -0.6275
kp    E=  1.6120 px=  1.2260 py= -1.0467 pz=  0.0000
pp    E=  1.2346 px=  0.0000 py=  0.8028 pz=  0.0000
qout  E=  1.2500 px= -1.2260 py=  0.2439 pz=  0.0000
q     E=  0.3289 px= -3.0274 py=  1.4050 pz=  0.6275
```